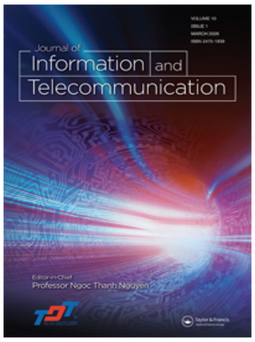

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A comprehensive survey on customer churn analysis studies

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ABSTRACT

This paper presents a comprehensive survey of customer churn analysis studies. It begins by examining the customer journey and touchpoints, emphasizing their influence on churn behaviour. The survey then reviews a range of machine learning algorithms used for churn prediction and explores the various feature types employed in this domain. Additionally, it investigates churn analysis across key industries, including telecommunications, banking, and gaming, highlighting the modelling approaches and features commonly applied within each sector. Finally, the paper identifies existing limitations in current research and proposes future directions to address these gaps and associated challenges. Overall, this survey aims to provide a thorough understanding of current practices and to support further advancements in the field of customer churn analysis.

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KEYWORDS
Customer churn; machine learning analysis; business domains; feature engineering; predictive models

1. Introduction

Regardless of the industry or product, customers are the most important component of a business's success and growth. Businesses make significant efforts to acquire, and more importantly, retain their existing customers. Customer satisfaction is directly linked to revenue growth, business credibility, and reputation, while customer dissatisfaction leads to churn and loss of revenue.

Customer churn is commonly used to describe a customer's tendency to stop doing business with a company within a given time-frame or contract (Tueanrat et al., 2021). Many companies encounter customer churn, in part because they lack effective tools to deliver a customer experience that sustains their competitive edge. Without adequate data analytics capabilities, these businesses face difficulties in fully understanding their customers. This limited visibility into customer behaviour contributes to weakened customer relationships and makes it difficult to anticipate when churn is likely to occur (Dias & Antonio, 2025).

Given the marketing and operational costs of customer acquisition and satisfaction, losing a customer to a competitor can be costly. Generally, it's less expensive to retain existing customers. Therefore, it's crucial for businesses to understand why and when a

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customer might stop using their services or switch to a competitor. This understanding enables proactive measures, such as offering incentives that encourage customers to remain, helping create a more positive customer experience.

Customer experience is a dynamic process, defined as the customer's journey with a firm over time during the purchase cycle across multiple touchpoints (Lemon & Verhoef, 2016). The customer journey, encompassing all interactions a consumer has with a company across various touchpoints, significantly influences their overall satisfaction with the offering. This experience, in turn, shapes long-term customer outcomes such as loyalty, success, and the likelihood of churn (Kuehnl et al., 2019).

Customer churn studies have become increasingly popular across various service sectors and businesses (e.g. telecom, insurance, banking, internet services, gaming [Ahn et al., 2020]), reflecting its importance. Each business has distinctive features and characteristics, leading to separate investigations of churn within each industry using various metrics. Consequently, a variety of analytical methods have been employed, tailored to the unique features of each business sector.

This study explores existing customer churn research from multiple perspectives, including customer journeys, touchpoints, feature engineering approaches, and applied machine learning techniques. It considers churn analysis across various business contexts, highlights current limitations, and proposes directions for future research to advance the field.

2. Customer journey and touchpoints

Having introduced the importance of retaining customers and the challenges in understanding churn, we now turn to a foundational concept that shapes customer satisfaction and churn, the customer journey and its associated touchpoints. Customer journey covers the entirety of a customer's experience with a brand. A brand name is a name which is used to identify a single line or the family of products or services offered to customers. For example, Nike is the brand name used on most products manufactured by Nike incorporation. On the other hand, Apple is a company with many brands such as iPad, iPhone, iPod, and Mac.

The customer journey encompasses both the procedural and experiential aspects of customer service. It is defined by the recurring interactions between a service provider and a customer. Often, the customer journey view is used to understand and design for customer experience. Adopting this perspective is crucial for involving customers in strategic planning, business model development, and service design (Følstad & Kvale, 2018). As illustrated in Figure 1, the customer journey is a process that consists of three sub-processes: 'Pre-Purchase', 'Purchase', and 'Post-Purchase' (Lemon & Verhoef, 2016). Each sub-process includes specific phases that reflect customer behaviours at that stage.

- **Pre-Purchase**: This phase involves a customer's engagement with a brand before making a purchase, including need recognition, search, and consideration.
- **Purchase**: This phase encompasses interactions with the brand during the actual transaction, such as selecting, ordering, and paying for a product or service. Although brief, it is critical as customers decide whether to proceed with the purchase among various touchpoints.

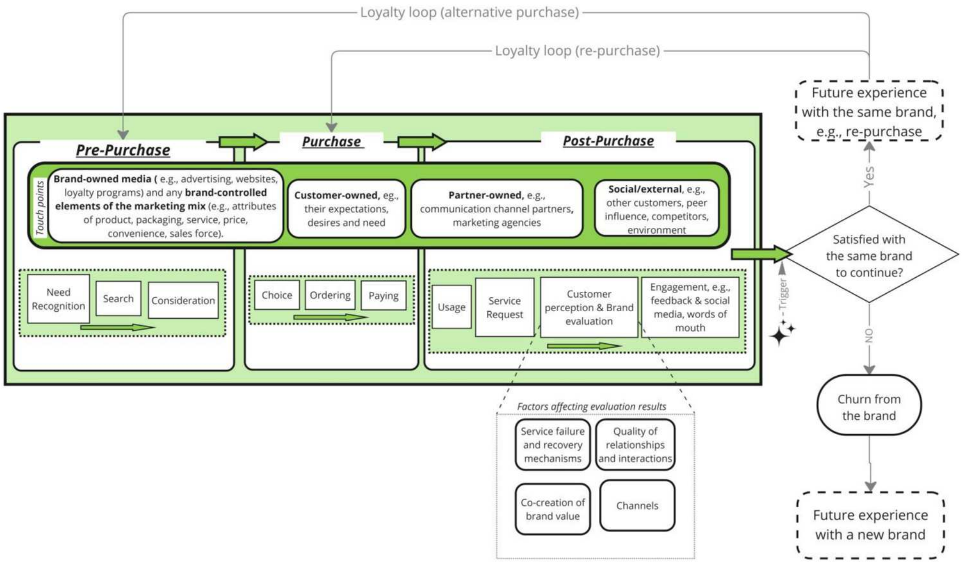

Figure 1. Customer journey process.

- **Post-Purchase**: This phase covers the customer's experience after the purchase, including product usage, engagement, and handling service issues. It can theoretically extend until the end of the customer's life.

Several managerial studies have expanded the post-purchase phase to include the 'loyalty loop' as part of the overall customer journey. These studies suggest that during the post-purchase phase, a trigger may either result in customer loyalty (through repurchase and further engagement) or cause the customer to return to the pre-purchase phase and reevaluate alternatives (Court et al., 2009).

In the customer journey, touchpoints refer to service or product interactions that a customer engages with so that either directly or indirectly, related to a specific brand. Touchpoints influence customer's views and assessments of the brand in general and the customer journey in particular (Towers & Towers, 2021; Tueanrat et al., 2021). These touchpoints occur across various channels, e.g. advertising (a touchpoint) through radio (a channel), and can be categorized into four types, namely, brand-owned, partner-owned, customer-owned, and social/external (Lemon & Verhoef, 2016).

- **Brand-owned Touchpoints** are those interactions that take place throughout the journey and are planned, handled, and controlled by the company. They consist of all media owned by the brand, including advertisements (Hanssens et al., 2014; Baxendale et al., 2015), web-pages, and loyalty programmes, as well as any marketing mix components (e.g. store design, attributes of product, packaging, service, price, convenience, sales force) that are under the control of a brand (Lemon & Verhoef, 2016).
- **Partner-owned Touchpoints** are planned, handled, or managed jointly by the company and a number of its partners. Marketing companies, multichannel distributors, and channel partners for communication are examples of partners.

- **Customer-owned Touchpoints** are customer behaviours that are a portion of the total customer experience however the company, its partners, or third parties have no control over them. For example, when buyers consider their own requirements or preferences prior to making a purchase. The customer's payment option during a purchase is largely a customer-owned touchpoint, however, partners may also be involved. The most important and common customer-owned touchpoints occur post-purchase when personal usage acts as a focal point.
- **social/external Touchpoints** express the significant roles of external factors in shaping the overall customer experience. Throughout the entire customer journey process, customers are exposed to external factors, such as other customers, peer influences, independent sources of information, and their surroundings, that can impact their experience. Peers can influence customers in all three stages of the journey, whether or not their input is solicited. Other customers can also influence them, through their behaviour or simply their presence, particularly during the purchasing process or when the product or service is consumed immediately after purchase, such as at restaurants, sporting events, theatres, or concerts (Towers & Towers, 2021).

  The impact of social/external touchpoints on customers can be considerable and in some cases, even more significant than the effect of advertising (Baxendale et al., 2015; Towers & Towers, 2021). Studies suggest that external sources of information, such as social media (De Vries et al., 2012) and review websites like TripAdvisor, as well as the social setting (Lin & Liang, 2011), can influence customers' choices. These sources of information may have varied degrees of association with the brand or company and may also be seen as points of contact with partners (Manchanda et al., 2015).

While businesses can control many touchpoints, such as advertisements, product attributes, and store design, they must also manage uncontrollable touchpoints like customer feedback and press coverage. Effective management of controllable touchpoints ensures positive customer experiences and can lead to favourable reviews and media coverage. However, a gap remains in the literature regarding how businesses can systematically integrate insights from uncontrollable or social/external touchpoints into churn mitigation strategies, despite their significant influence.

Customer services, which include communication and follow-up after sales, provide valuable insights into customer sentiment. By optimizing touchpoints and integrating preferred communication channels into their strategy, businesses can enhance customer satisfaction and gather essential market feedback for strategic decisions. While all these categories of touchpoints provide a comprehensive framework for understanding customer-brand interactions, limited research has investigated how dynamic customer preferences alter the impact of each touchpoint over time. Further empirical studies are needed to examine how context-driven factors modulate touchpoint effectiveness.

2.1. Customer churn dimensions

Having established the importance of the customer journey and touchpoints, we now examine the various dimensions and definitions of churn, which vary depending on contractual settings, industry context, and behavioural assumptions. Generally, a dictionary-based definition of churn is inactivity for a long time (Perianez et al., 2016). However, each business could have its own definition of 'long time' and 'inactivity'. This variety in definition arises from the intense market competition and non-strict membership contracts and conditions in recent services offered to the customers. For example, in the industry of internet services (i.e. businesses that provide various services and products over the internet such as online Banking and Financial Service, online Learning and education, gaming and Online Entertainment), recurrent customer churn is happening because of the reduced investment expenses for customers (Buckinx & Van den Poel, 2005; Tamaddoni Jahromi et al., 2010), and slight switching costs when changing services (Lejeune, 2001).

The earliest research on churn was done from a management standpoint. Especially Customer Relationship Management (CRM) has historically been the starting point for studies on customer attrition (Ngai et al., 2009). CRM is a managerial method, aiming at increasing the efficiency of businesses (Parvatiyar & Sheth, 2001).

CRM is a two-part system that includes both operational and analytical CRM. Analytical CRM specifically concentrates on customers' behaviour, personal traits, and attitudes to generate tailored marketing strategies and meet the needs of individual customers (Ngai et al., 2009).

With churn analysis emerging as a vital personalized customer management method, IT-related technologies are used in analytical CRM, and businesses are implementing those technology-derived solutions for customer selection, acquisition, and retention (Ahn et al., 2020). Hence, customer churn prediction and analysis are vastly used as crucial personalized methods in the customer management field (Tamaddoni Jahromi et al., 2010). Customer churn analysis have been expanded in different industries and domains such as telecommunications, finance, gaming, insurance, retail, internet services and e-commerce (Nagaraj et al., 2023), media streaming.

A timeline of customer churn analysis is depicted in Figure 2. The timeline begins with Customer Lifetime Value (CLV) prediction models, which incorporate calculations of customer profitability and churn rates (Morrison & Schmittlein, 1988). CLV models help businesses forecast the total value a customer will bring over their lifetime, influencing retention strategies. This is followed by Traditional Machine Learning approaches (Mozer et al., 2000). These methods, including decision trees, logistic regression, and support vector machines, use historical data to predict churn by identifying patterns and relationships in customer behaviour. Statistical methods, such as Negative Binomial

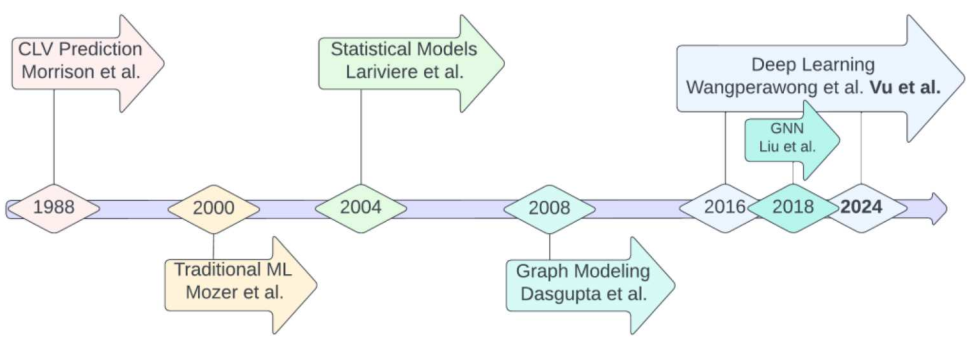

Figure 2. Timeline of customer churn analysis.

Distributions (NBD), are also used for CLV prediction. NBD models account for the variability in customer purchase patterns and are particularly useful for predicting customer spending and frequency.

Statistical models shown in the figure represent the early adoption of literature where models built with statistical methods were used to analyse customer churn (Lariviere & Van den Poel, 2004). These models provide insights into churn patterns by evaluating customer behaviour data through statistical techniques.

Initially, graph modelling approaches (Dasgupta et al., 2008) were applied in the telecom domain. These approaches map customer interactions and relationships to analyse and predict churn based on network structures and social connections. Later, with the rise of Deep Learning (Vu, 2024), more advanced graph analytics using neural networks became feasible (Liu et al., 2018). Deep learning models, such as convolutional neural networks (CNN) and recurrent neural networks (RNN), can capture complex patterns in large datasets and improve the accuracy of churn predictions by learning from more detailed features and interactions.

Customer churn could be viewed from two standpoints, namely involuntary and voluntary churn. Involuntary churn occurs when customers misuse services or fail to pay their bills or debts. On the other hand, voluntary churn is driven by customers' own choices and can be challenging to identify. Incidental churn typically accounts for a small portion of a company's overall voluntary churn. In those situations, there are factors beyond the company's control influencing customer decisions. Incidental churn occurs when customers no longer need a service due to changes in their circumstances. This can be triggered by financial constraints or relocating to an area where the company's service is not available. Examples include customers being unable to afford the service or moving to a different location (Hadden et al., 2008). On the other hand, Coussement et al. classified churn-related risks into 'Controllable', 'Uncontrollable', and 'Unknown'. The Controllable risk includes factors like pricing and service quality, while the Uncontrollable risk includes factors like migration or death. The Unknown risk contains customers who did not provide a reason for churning (Coussement & Van den Poel, 2008).

Churn could also be defined by contract-related criteria, i.e. being a contract-based churn or non-contract-based churn (Ahn et al., 2020). Churn is an event in which a customer terminates a relationship with a business at some point. For instance, the termination of the relationship could be in the form of a bank account closure while in an online-based service, churn could be an event in which a customer stops purchasing products and services from its website. The former relationship is called a contractual setting in which a customer explicitly informs a business about its churn by notifying the account closure. In the latter case, known as a non contractual setting, churn prediction is more complex, as it is latent because a customer can silently terminate its relationship with a business. Therefore, a prerequisite for churn studies is to determine whether the relationship between a business and a customer is contractual or non-contractual. This influences the definition of churn and, eventually, a modelling approach (Hasumoto & Goto, 2022).

In addition to contract-related criteria, churn could be considered from three aspects, i.e. monthly churn, daily churn, or binary churn. Both monthly and daily churn are linked to when the customer's information is updated in the database of a company. The binary churn observation is obtained by applying the database information in line with the managers' analytical targets. Essentially, binary churn is determined by the presence of a contract in the contractual settings. In non- contractual situations, the company sets measures for customer inactivity. When a customer exhibits those inactive behaviours, they are considered binary churn (Buckinx & Van den Poel, 2005).

There are numerous previous churn studies, conducted on contractual settings with monthly churn measures (Chen et al., 2018; Dahiya & Bhatia, 2015; Bahnsen et al., 2015). These monthly data-based Churn analysis miss out daily insights. They often identify churn too late for effective intervention. Additionally, they face challenges in handling the high volume of data when considering daily customer behaviour (Alboukaey et al., 2020). On the other hand, most of the studies that have been conducted on non-contractual settings such as game industry (Yang et al., 2020; Hadiji et al., 2014; Runge et al., 2014), social online platforms (Yang et al., 2018), and Pre-paid mobile telephony[1] (Tamaddoni Jahromi et al., 2010) applied daily churn data on their analysis. There are also several studies on the contractual settings with binary churn measures (Zhang et al., 2017), while some considered daily churn on contractual settings, such as telecommunication (Alboukaey et al., 2020). The growing diversification of churn definitions across industries introduces modelling inconsistencies. An unresolved challenge remains in developing adaptable churn models that maintain predictive power across both contractual and non-contractual environments.

3. Churn analysis: techniques and evaluation metrics

Most churn-related studies are done with the goal of identifying or predicting the propensity of customers to leave a company and not use its services any more. The churn rate, also known as the attrition rate, is mostly used for churn and retention calculations. Churn rate refers to the ratio of the number of lost customers, who stopped using a product or a service, to the total number of customers within a specific period (Wei & Chiu, 2002). As such, churn prediction can be used as a method to help with decreasing the churn rate, increasing the retention rate of loyal customers, and ultimately causing an increment in the value of the company (Ahn et al., 2020). We present a hierarchical diagram, Figure 3 that organizes customer churn analysis studies along four key dimensions, i.e. industries, feature types, model types, and evaluation metrics. Customer churn poses a significant threat to firms operating on a subscription-based model, including industries such as insurance (Gunther et al., 2014), banking (Anil Kumar & Ravi, 2008), online gambling (Coussement & De Bock, 2013), online video games (Kawale et al., 2009), music streaming (Chen et al., 2018), online services (Tan et al., 2018), and telecommunication (Hosein et al., 2021). These businesses rely on consistent and predetermined membership fees, making customer retention crucial for sustainable profitability. In response to this challenge, accurately predicting customer churn has become a top priority in these industries. Beyond simply identifying customers who are likely to switch, companies also need to understand the underlying causes of churn behaviour. This dual approach not only helps with profiling customers prone to churn but also enables the implementation of effective proactive retention campaigns (Leung et al., 2021). Tables 1 and 2, present various previous churn studies across different domains, along with their respective publication venues.

Once features are defined and data is collected, the next critical step is selecting an appropriate modelling technique. This section discusses the main modelling approaches

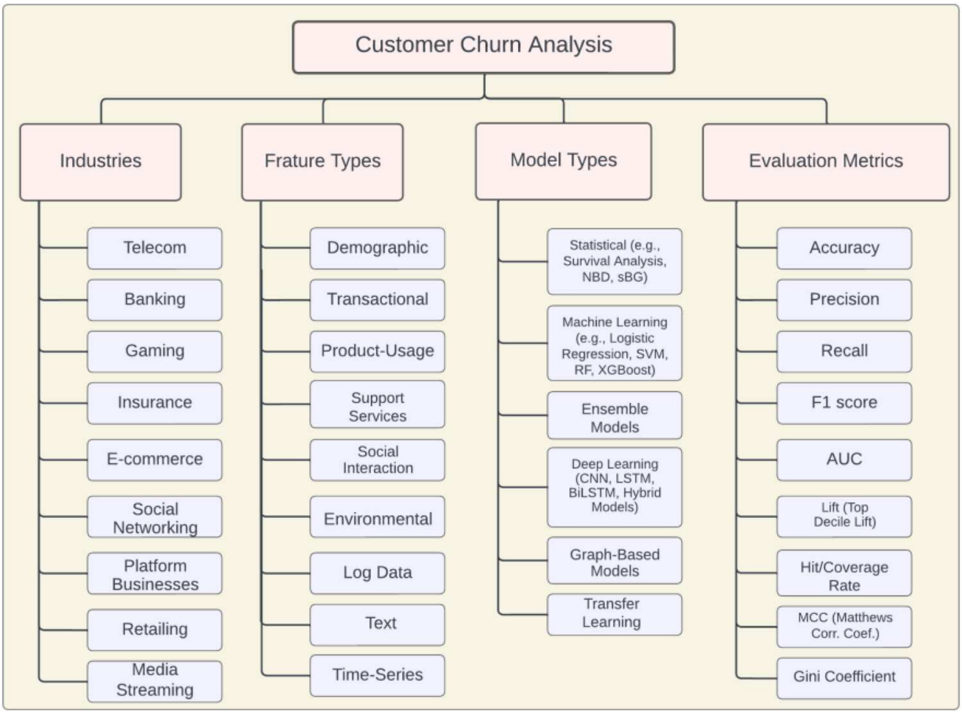

Figure 3. A hierarchical taxonomy of customer churn analysis dimensions. This diagram categorizes churn studies based on four key dimensions: industries, feature types, model types, and evaluation metrics. It reflects the diversity of methodological choices and contextual applications discussed throughout this survey.

for predicting customer churn, including traditional machine learning and emerging deep learning methods. Machine learning methods have evolved alongside the development of data mining since the advent of computers, while statistical methods have primarily focused on hypothesis testing based on mathematical principles (Witten & Frank, 2002).

Modelling techniques vary depending on the business domain, and the choice of approach is a crucial aspect of churn analysis. The most suitable algorithm depends on the decision maker's objectives (Huang et al., 2012). There is a growing trend in industries like gaming and telecommunications, which have abundant log data and easy access to customer information, to increasingly adopt deep learning with big data. In contrast, in sectors like finance and insurance, where log data is limited and customer information changes less frequently, statistical methods and traditional machine learning models, such as survival analysis, are more commonly used (Ahn et al., 2020).

3.1. Statistical approach

In the field of statistics, probability models have been widely applied to predict customer churn, particularly in the analysis of customer bases (Fader & Hardie, 2009). In these analyses, churn rates play a crucial role in estimating survival time, which is essential for calculating CLV. These prediction algorithms integrate expected customer revenue with models for predicting churn rates.

Table 1. Previous churn studies on different domains and the corresponding publication venues.

Year	Domain	Publication venue	Ranking	Cite	Ref.
2025	Gaming	IEEE Transactions on Consumer Electronics	H-Index: 2.5/Q1	0	Li et al. (2025)
2025	Telecom	Algorithms	H-Index: 58/Q2	0	Shahabikargar et al. (2025)
2025	Telecom	Results in Engineering	H-Index:56/Q1	1	Asif et al. (2025)
2025	Telecom	ICISCN 2025 (IEEE)	Rank: B	0	Nimma et al. (2025)
2024	Multiple Industries	Technological Forecasting and Social Change	H-Index:209/Q1	0	Liu, Jiang, et al. (2024)
2024	Telecom	Results in control and optimisation	H-index:12/Q2	57	Wagh et al. (2024)
2024	Telecom	Kybernetes	H-Index: 59/Q2	12	Subramanian et al. (2024)
2024	Telecom	Journal of Big Data	H-Index:91/Q1	0	Sagming et al. (2024)
2024	Telecom	IEEE Access	H-index:204/Q1	0	Saha et al. (2024)
2024	Telecom	Expert systems with applications	H-index:249/Q1	0	Soltani et al. (2024)
2024	Banking	Neural Computing and Applications	H-Index: 146/Q1	11	Vu (2024)
2023	E-Commerce	Measurement: Sensors	H-Index:33/Q2	56	Shobana et al. (2023)
2023	Telecom	Scientific Reports (Nature)	H-Index:374/Q1	37	Khattak et al. (2023)
2023	Banking and Insurance	Neural Computing and Applications	H-index:111/Q1	21	Kate et al. (2023)
2023	Gaming	Central European Journal of Operations Research	H-index: 38 / Q3	2	Periˇsi´c and Pahor (2023)
2023	Gaming	Big Data	H-index: 45 / Q2	6	Weiss and Vilenchik (2023)
2023	Telecom	Sustainability	H-index: 136 / Q2	15	Saha et al. (2023)
2023	Telecom	Applied Soft Computing	H-Index:208/Q1	57	Amin et al. (2023)
2022	Banking	Data Journal	IF: 2.6	10	Bharathi et al. (2022)
2022	Social Networking	ACM Int. Conf. on Web Search and Data Mining	Rank: A	5	Zhang et al. (2022)
2022	Platform Businesses	Neural Computing and Applications	H-index: 111 / Q1	8	Hasumoto and Goto (2022)
2022	Telecom	Computing Journal	H-index: 64	186	Lalwani et al. (2022)
2021	Telecom	Journal of Big Data	H-index: 61 / Q1	3	Colot et al. (2021)
2020	Retailing	European Journal of Operational Research	H-index: 288 / Q1	200	Martinez et al. (2020)
2020	Banking	Journal of Physics: Conference Series	H-index: 91 / IF: 3.7	3	Li and Xie (2020)
2020	Telecom	Expert Systems with Applications Journal	H-index: 249	80	Alboukaey et al. (2020)
2019	Telecom	Journal of Big Data	H-index: 61 / Q1	385	Ahmad et al. (2019)
2019	Gaming	IEEE Conference on Games (CoG)	Rank: C	22	Kristensen and Burelli (2019)
2018	Gaming	IEEE / Data Mining (ICDM)	Rank: A*	44	Liu et al. (2018)
2018	Gaming	Machine Learning and Knowledge Extraction	IF: 3.9	11	Guitart et al. (2018)
2018	Gaming	IEEE Transactions on Games	H-index: 48 / Q3	40	Lee, Kim, et al. (2018)
2018	Social Networking	ACM Int. Conf. on Knowledge Discovery & Data Mining (KDD)	Rank: A*	99	Yang et al. (2018)
2018	Telecom	European Journal of Operational Research	H-index: 288	105	Hoppner et al. (2020)
2018	Telecom	IEEE / Smart Computing and Electronic Enterprise (ICSCEE)	H-index: 11	47	Agrawal et al. (2018)
2017	Telecom	Journal of Engineering and Technology (IRJET)	H-index: 11	64	Umayaparvathi and Iyakutti (2017)
2017	Telecom	IEEE / Computational Intelligence and Computing Research (ICCIC)	H-index: 11	49	Mishra and Reddy (2017)
2017	Gaming	Expert Systems with Applications	H-index: 249 / IF: 8.5	103	Milosevic et al. (2017)
2017	Gaming	PLOS ONE Journal	H-index: 404 / IF: 3.7	79	Kim et al. (2017)
2017	Gaming	IEEE Conference on Computational Intelligence and Games (CIG)	Rank: C	12	Jeon et al. (2017)

(*Continued*)

Table 1. Continued.

Year	Domain	Publication venue	Ranking	Cite	Ref.
2017	Gaming	IEEE Conference on Computational Intelligence and Games (CIG)	Rank: C	67	Bertens et al. (2017)
2017	Insurance	IEEE / Services Computing	Rank: B	50	Zhang et al. (2017)
2016	Retailing	Journal of Service Research	H-index: 137 / Q1	51	Tamaddoni et al. (2016)
2016	Gaming	IEEE / Data Science and Advanced Analytics (DSAA)	Rank: B	116	Perianez et al. (2016)
2016	Gaming	IEEE Conference on Computational Intelligence and Games (CIG)	Rank: C	34	Sifa et al. (2016)
2016	Gaming	AAAI Conf. on Artificial Intelligence and Interactive Digital Entertainment	H-index: 14	49	Drachen et al. (2016)
2016	Gaming	IEEE Conference on Computational Intelligence and Games (CIG)	Rank: C	49	Tamassia et al. (2016)

For contractual settings, the shifted-Beta-Geometric (sBG) model is commonly used. This model employs a beta distribution to account for shifts at each time instance (t), providing a suitable fit for the retention rate. As a result, the sBG model allows for a continuous interpretation of customer retention in a discretetime, contractual service environment (Fader & Hardie, 2007). In contrast, for non-contractual settings, recurrent purchasing patterns are often characterized using negative binomial distributions (NBD) (Tamaddoni et al., 2016).

Moreover, the churn distribution is modelled using a gamma mixture of exponential, also recognized as the Pareto (second kind) distribution. Integrating this with customer's behaviour and survival distribution enables the calculation of CLV, known as the Pareto/NBD model (Schmittlein et al., 1987). This method has been widely employed as a probability model for CLV derivation until recently, highlighting its continued relevance in understanding and predicting customer behaviour in business contexts (Fader et al., 2010). Alternatively, the beta-geometric/beta-binomial (BG/BB) model serves as another approach. In this model, the beta-geometric component aligns with the retention rate, while the beta-binomial distribution corresponds to modelling customer purchasing behavioural pattern (Fader & Hardie, 2009).

In various domains, particularly finance, statistical modelling methods are often employed to address churn analysis and prediction. Survival analysis is a popular method, used to model the happening and timing of events (Van den Poel & Lariviere, 2004; Mavri & Ioannou, 2008). In those analysis the time to failure considered to be related to the corresponding customer churn behaviour. Additionally, structural equation modelling has been used to analyse potential churner behaviour, evaluating the effect of suspected influential features on customer decisions such as churn (Varki & Colgate, 2001). Exploring variance (Zeithaml et al., 1996), T-tests and Chi-square statistics (Hitt & Frei, 2002), Kaplan-Miere model (Lariviere & Van den Poel, 2004), Cox proportional hazards models (Ali & Arıturk, 2014; Viljanen et al., 2016), and A/B tests (Runge et al., 2014; Milosevic et al., 2017) have also been widely used in financial areas to getting some insights into customer behaviour and perceptions.

In non-contractual scenarios, customer churn exhibits a continuous probabilistic nature. Defining and tracking customer churn in these settings is challenging. Researchers employ time-series modification techniques, such as grouping customer IDs for data organization or calculating behavioural variances, to define customer churn more

Table 2. Previous churn studies on different domains and the corresponding publication venues (continue).

Year	Domain	Publication venue	Ranking	Cite	Ref.
2015	Gaming	IEEE Conference on Computational Intelligence and Games (CIG)	Rank: C	69	Xie et al. (2015)
2015	Gaming	AAAI Conf. on Artificial Intelligence and Interactive Digital Entertainment	H-index: 14	134	Sifa et al. (2015)
2015	Gaming	SAI Intelligent Systems Conference (IntelliSys)	H-index: 14	42	Rothenbuehler et al. (2015)
2015	Banking and Insurance	Engineering Applications of Artificial Intelligence Journal	H-index: 126 / IF: 7.8	151	Sundarkumar and Ravi (2015)
2015	Telecom	European Conference on Information Systems (ECIS)	H-index: 22	27	Baumann et al. (2015)
2015	Telecom	IEEE / Reliability, Infocom Technologies and Optimization (ICRITO)	H-index: 15	123	Dahiya and Bhatia (2015)
2015	Telecom	Decision Analytics	H-index: 3	67	Bahnsen et al. (2015)
2015	Telecom	Proceedings of the AAAI Conference on Artificial Intelligence	Rank: A*	21	Amiri and Daume (2015)
2015	Telecom	Simulation Modelling Practice and Theory Journal	H-index: 78	553	Vafeiadis et al. (2015)
2014	Gaming	IEEE Conf. on Computational Intelligence and Games	Rank: C	211	Runge et al. (2014)
2014	Retailing	Industrial Marketing Management	H-index: 161 / Q1	191	Jahromi et al. (2014)
2014	Gaming	ACM Conf. on Computer Supported Cooperative Work & Social Computing	H-index: 48	162	Grandprey-Shores et al. (2014)
2014	Gaming	IEEE Conf. on Computational Intelligence and Games	Rank: C	221	Hadiji et al. (2014)
2014	Insurance	Scandinavian Actuarial Journal	H-index: 44	78	Gunther et al. (2014)
2014	Banking	Expert Systems with Applications	H-index: 249 / IF: 8.7	149	Ali and Arıturk (2014)
2014	Telecom	Expert Systems with Applications Journal	H-index: 249	112	Kim et al. (2014)
2014	Telecom	Applied Soft Computing Journal	H-index: 171	191	Verbeke et al. (2014)
2014	Telecom	Intelligent Data Analysis Journal	H-index: 50	50	Verbraken et al. (2014)
2012	Telecom	Expert Systems with Applications Journal	H-index: 249	428	Huang et al. (2012)
2012	Telecom	European Journal of Operational Research	H-index: 288 / Q1	551	Verbeke et al. (2012)
2012	Insurance	Intl. Proc. of Computer Science and Information Technology	H-index: 5	37	Soeini and Rodpysh (2012)
2010	Telecom	Journal of Strategic Marketing	H-index: 61 / Q2	70	Tamaddoni Jahromi et al. (2010)
2010	Telecom and Insurance	Journal of Interactive Marketing	H-index: 120	111	Risselada et al. (2010)
2009	Banking	Expert Systems with Applications	H-index: 249 / IF: 8.6	536	Xie et al. (2009)
2008	Banking	Managerial Finance	H-index: 45 / Q3	100	Mavri and Ioannou (2008)
2008	Telecom	Intl. Journal of Industrial and Manufacturing Engineering	H-index: 8	117	Hadden et al. (2008)
2008	Telecom	Journal of Extending Database Technology	H-index: 23	496	Dasgupta et al. (2008)
2007	Telecom	Knowledge-Based Systems Journal	H-index: 151 / Q1	174	Chu et al. (2007)
2006	Telecom	Expert Systems with Applications	H-index: 249 / Q1	810	Hung et al. (2006)
2006	Banking	Decision Support Systems	H-index: 170 / Q1	92	Prinzie and Van den Poel (2006)
2004	Banking	Expert Systems with Applications	H-index: 249 / IF: 8.5	177	Lariviere and Van den Poel (2004)
2004	Banking	European Journal of Operational Research	H-index: 288 / Q1	561	Van den Poel and Lariviere (2004)
2003	Banking	Expert Systems with Applications	H-index: 249 / IF: 8.5	143	Chiang et al. (2003)
2003	Telecom	IEEE Transactions on Evolutionary Computation	H-index: 199 / Q1	502	Au et al. (2003)
2000	Telecom	IEEE Transactions on Neural Networks	H-index: 234 / Q2	517	Mozer et al. (2000)

effectively in non-contractual situations (Runge et al., 2014; Milosevic et al., 2017). Researchers initiate the prediction of customer churn by leveraging features derived from the processed data. The foundation of employed statistical models are statistical inference, hypothesis testing, while survival analysis with hazard methods are utilized to construct effective churn prediction models (Lariviere & Van den Poel, 2004).

3.2. Machine learning approach

Machine learning (*ML*) techniques have been increasingly applied to address customer churn problems (Geiler et al., 2022). Unlike traditional statistical methods, machine learning exhibits robust capabilities in capturing non-linear relationships among features, as well as effectively learning heterogeneous effects from different feature sets (Ahn et al., 2020). Among the ML algorithms that are frequently used in the field of churn prediction are K-nearest neighbours, Naive Bayes, Logistic Regression, Support Vector Machines, Bagging, and Boosting classifiers.

k-nearest neighbours: The k-nearest neighbours (k-NN) algorithm is a non- parametric memory-based method. It determines the label for an instance x_i by selecting the most frequently occurring label among its k nearest training samples. K-NN is non-parametric because it doesn't make explicit assumptions about the functional form of the underlying data distribution. It doesn't assume a specific structure for the model, which provides flexibility and adaptability to different types of data. It is also memory-based because it 'memorizes' the entire training dataset. During the prediction phase, it refers to the training data to make predictions based on the similarity between new instances and instances in the training set. For a given instance x_i, which is a data point in the feature space, the algorithm assigns a label based on the majority label among its k closest training samples. In other words, it looks at the k training samples that are most similar or 'nearest' to x_i. The algorithm uses a majority voting rule to determine the label for the instance x_i.

Among the k nearest neighbours of x_i, it counts the occurrences of each class label and assigns the label that is most frequent (i.e. has the majority).

The ease of use and effectiveness of k-NN have made this method quite popular in the machine learning area. Different previous studies leveraged KNN in their online gaming churn problem analysis (Castro & Tsuzuki, 2015). Nevertheless, when used on data that resembles churn patterns, it has several noticeable limitations (Ahn et al., 2020).

Naive Bayes classifier: A Naive Bayes classifier computes the probability that a given input sample belongs to a specific class. The general term Naive Bayes refers to the strong independence assumptions in the model, rather than the particular distribution of each feature. The multinomial Naive Bayes classifier is a specific implementation that employs a multinomial distribution for each feature. This particular variant is well-suited for handling discrete data, making it effective in scenarios where features represent counts or frequencies. On the other hand, the Gaussian Naive Bayes (Gnb) classifier assumes that feature values adhere to a Gaussian (normal) distribution, making it suitable for continuous data.

In Problems where density estimation is challenging, the Gnb classifier becomes well-suited, especially in high-dimensional feature spaces. This classifier, based on the Naive Bayes framework, assumes that the conditional probabilities of features given the class label follow a Gaussian (normal) distribution. Despite its simplicity and the 'naive' assumption of feature independence, Gnb can be effective in situations where the distribution of features is approximately Gaussian. This characteristic makes it particularly valuable in handling high-dimensional data, where more complex models may face challenges in estimating densities. Additionally, it allows for efficient and computationally feasible classification, providing a pragmatic solution for certain machine learning applications (John & Langley, 2013). While the Naive Bayes classifier may exhibit sensitivity to the class imbalance issue (Bermejo et al., 2011), it still demonstrates the capability to yield favourable results for churn prediction tasks (Huang et al., 2012).

In their churn prediction research, Verbraken et al. demonstrated that Bayesian Network classifiers do not surpass the performance of the logistic regression. Nevertheless, their valuable contribution stems from providing a highly intuitive understanding of the relationships among the explanatory variables. This offers insights into dependencies that may enhance interpretability (Verbraken et al., 2014).

Logistic regression (LR): LR predicts the conditional probability of the classes, by using a linear function. In scenarios involving imbalanced datasets, research indicates that the bias in the regression vector intercept tends to be more significant when the class distribution is skewed (Tamaddoni et al., 2016). This challenge can be addressed through a prior adjustment that considers the minority class. Alternatively, it can be tackled by employing a penalized likelihood approach. In the latter method, the maximum likelihood formula is modified by assigning weights based on the proportion of instances belonging to the positive class in the target variable. This adjustment ensures that the model is more responsive to the minority class, emphasizing increased sensitivity to better capture its characteristics in the dataset. Techniques like these are valuable for improving the performance of models, especially in scenarios with imbalanced class distributions (Kristensen & Burelli, 2019). The effective performance of logistic regression (LR) has been previously highlighted in previous churn prediction studies (Burez & Van den Poel, 2009), including churn in telecom (Huang et al., 2012; Verbraken et al., 2014), banking (Ali & Arıturk, 2014), multiplayer games (Grandprey-Shores et al., 2014), insurance (Risselada et al., 2010; Gunther et al., 2014), retailing (Migueis et al., 2012), and loyalty card company (Lim et al., 2017). Although logistic regression is widely used and easy to interpret, its linear nature limits its ability to model complex interactions. This shortcoming has accelerated the shift towards non-linear models like ensemble methods and deep learning in high-dimensional churn scenarios.

Support Vector Machine: The Support Vector Machine (SVM), serving as a kernel-based *ML* model for classification and regression tasks. The SVM classifier is designed to find a hyperplane that optimally separates classes, prioritizing a clear margin between them. Its flexibility allows it to handle both linearly separable and non-separable cases, making it a versatile tool for various classification problems. The degree of adaptability to non-separable cases is controlled by tuning parameters during the SVM training process. Support Vector Machines have been used in several studies on customer churn analysis. Promising results have emerged from these research in a variety of industries, including the energy sector (Moeyersoms & Martens, 2015), online gaming (Xie et al., 2015), and telecommunication (Huang et al., 2012; Vafeiadis et al., 2015).

Bagging: The Bagging (Bootstrap Aggregating) is an ensemble technique designed to enhance the performance of unsteady estimation or classification models (Risselada et al., 2010). It aims to decrease variance by training base learners such as decision trees, independently, on random subsets of the training data. This technique stands out from fundamental ensemble algorithms. It achieves this distinction by creating a new model using a bootstrap resample of a size less than *n*. In the process of training *M* models, the final decision $f^{\wedge}bag$ is determined. This decision is reached by averaging the *M* decision rules $f^{\wedge}m(X)$ obtained from the bootstrapped training sets. Random Forest also utilizes bagging on decision trees, in the process of variable sampling as part of its methodology.

The *DT* algorithm creates sub-partitions by selecting a variable from the available features and splitting the data based on an impurity criterion, such as *Gini*. *DT* is known for its easily interpretable decision rules. In contrast, *RF* selects variables within a random subset of features for each split in each tree. It is an ensemble of multiple decision trees, and this ensemble strategy often leads to more accurate predictions compared to a single Decision Tree. However, the decision rules in Random Forest are not as easily interpretable as those of a single Decision Tree due to the complexity introduced by multiple trees.

Despite the reduced interpretability, Random Forest has an advantage in providing a measure of feature importance. It can indicate which features are more influential in contributing to the overall accuracy of the model. On the other hand, previous studies have highlighted the good performance of Random Forest for feature selection on imbalanced datasets, with uneven class distribution, as observed in studies related to customer churn (Li & Xie, 2020; Rahman & Kumar, 2020). Sifa et al.'s study demonstrated a promising result when Applying *RF* technique while dealing with customer churn problem in a mobile online gaming field (Sifa et al., 2015). Similarly, employing *RF* in another study, addressing churn issue on online gaming, showcased encouraging outcomes (Lee, Kim, et al., 2018).

Boosting: The boosting method is identical to bagging in combining outcomes from various classifiers, typically decision trees. However, in the boosting strategy, each model aims to minimize errors from the preceding model, unlike bagging. Popular boosting variants include AdaBoost, gradient boosting, and stochastic gradient boosting. AdaBoost (Adaptive Boosting)'s distinct feature lies in utilizing observation weights w_i, higher for misclassified samples. Consequently, this approach compels the model $f^\wedge m$ to focus more on data for which it performs poorly, frequently updating weights. Each model strives to minimize the weighted error e_m, the sum of weights for misclassified observations.

XGBoost (eXtreme Gradient Boosting) is One of the best and most popular ways to use the Gradient Boosting algorithm. The core concepts of this supervised learning approach include function approximation, using different regularization approaches, and optimizing predefined loss functions. Regularization helps to prevent overfitting and improve the generalization ability of a model. XGBoost is designed to be highly efficient and support parallelization, making it faster than traditional gradient boosting. It has mechanisms to handle missing values during the training process (Chen & Guestrin, 2016).

3.3. Graph-based approach

Graph theory can be used for conceptualizing churn as a mathematical relationship. By configuring graph attributes based on features and individual customers, the relationship between them is expressed as edges within the graph. Once the graph structure is established, churning customers are identified through correlation analysis, leveraging the interconnected nature of the graph to reveal patterns indicative of potential churn.

In customer churn studies, researchers and analysts have employed graph theory methods to comprehensively examine the dynamics of customer relationships within a telecom network, using the CDR data (Kim et al., 2014; Ahmad et al., 2019). These methods include various relational models such as 'Class-Distribution Relational Neighbor classifier (CDRN)', 'Network-only Link Based classifier (NLB)', 'Spreading Activation Relational Classifier (SPA RC)', 'Weighted-Vote Relational Neighbor classifier (WVRN)', leveraged to perform analysis over social network connections (Verbeke et al., 2014). The interconnected nature of customers within the network is recognized as a fundamental aspect of the approach. A graph, in this context, is utilized to symbolize the relationships existing between customers, providing a visual representation of the network's structure (Kim et al., 2014). However, Liu et al addressed the churn problem in gaming industry, using graph neural networks and embeddings. In their approach, the relationships between players and games are illustrated using an attributed bipartite graph. The solid line represents an edge in the graph, where a node can be either of type player or game (Liu et al., 2018).

Graph theory is used to model and examine consumer relationships to explore their complexities. Finding trends or other elements that might help in predicting and reducing customer loss is the main goal. Analysing a graph entails evaluating its general structure as well as individual patterns within it. This examination helps to find clues or early warning signs that indicate possible churn issues (Calzada-Infante et al., 2020). A sudden disconnection of a core node is one example of such a signal that might point to a higher risk. Finally, to improve predictions, features obtained from the graph and traditional customer-related features are fed into machine learning algorithms. It would help with identifying behavioural patterns and make predictions about the likelihood of churn for each customer. Combining graph theory with machine learning could lead to create a robust strategy for predicting and managing customer churn in a telecom network (Verbeke et al., 2014; Kim et al., 2014; Amiri & Daume, 2015). Despite the promise of graph-based models, especially in telecommunications, their scalability and interpretability remain under-explored. Future studies should prioritize lightweight graph models that balance accuracy with business usability.

3.4. Deep learning approach

Deep learning methods have achieved state-of-the-art results across a range of application domains. Deep learning, an advanced extension of machine learning, employs neural network algorithms to delve into complex patterns in data. Despite sharing similarities, deep learning is often distinctively categorized from traditional machine learning due to its unique characteristics (Ahn et al., 2020). Deep learning can be applied to various types of machine learning tasks, whether supervised, unsupervised, or semi- supervised learning. The choice of the learning paradigm depends on the specific problem at hand and the availability of labelled data.

In the customer churn prediction area, deep learning techniques offer advanced methodologies. These are two leading strategies that are commonly employed in this context. The first approach involves training deep learning models using sparse customer data that undergoes a process called densification. Sparse data often contains gaps or lacks comprehensive information, making it challenging for traditional models. Densification aims to address this limitation by augmenting or expanding the dataset, filling in gaps to ensure a more comprehensive training set. This enables the model to grasp a more nuanced understanding of customer behaviour (Zhang, 2020). The second approach involves constructing a fully connected neural network by extracting latent vectors from Autoencoders and concatenating these features with static ones. This signifies a shift towards leveraging the power of deep learning to handle complex relationships within customer data, emphasizing its role as an evolving and effective tool in the domain of customer churn prediction (Ahn et al., 2020).

Recent studies (Shahabikargar et al., 2025; Spanoudes & Nguyen, 2017; Kristensen & Burelli, 2019) highlight the growing interest in applying deep learning methodologies in churn prediction. It is considered a subset of machine learning, representing a more recent analytical method in the field (Goodfellow et al., 2016). Although, its increasing academic popularity has come to be recognized as a separate academic discipline. Lee et al. revealed that a deep learning model outperformed a traditional machine learning model in predicting customer churn probability for game churn analysis. They used generalization and memorization approaches in their feature engineering process. Their approach, i.e. combining a deep learning model with a conventional machine learning model, resulted in improved predictability of churn (Lee, Jang, et al., 2018).

There are plenty of studies with promising results that applied deep learning techniques to address customer churn problem (Alboukaey et al., 2020). Those studies have been conducted in different domains, e.g. Guitart et al. (2018) used DNN in combination with LSTM, while Kristensen et al. (Kristensen & Burelli, 2019) applied LSTM for their churn analysis in online gaming domain. On the other hand, CNN has a great popularity in churn analysis in telecom industry (Umayaparvathi & Iyakutti, 2017; Mishra & Reddy, 2017). Various research is also done in financial field, such as banking (Wang et al., 2023) and insurance (Zhang et al., 2017), in which attention and Deep Shallow mixture Model (DSM) have been explored for getting favourable results on churn prediction problem. LSTM along with Attention, and CNN resulted in encouraging outcomes for churn analysis in social online platforms (Yang et al., 2018) and retail industry (Dingli et al., 2017), accordingly. Finally, it should be noted that although deep learning can enhance analytical performance, understanding its internal mechanism and outcomes which are crucial for businesses, remains challenging (Hasumoto & Goto, 2022).

Recent works have further advanced this space. For instance, hybrid deep learning models are proposed to combine CNN with Bidirectional Long Short-Term Memory (BiLSTM) to capture both spatial and sequential customer behaviour features. This architecture demonstrated significant improvements in identifying churn patterns, particularly in telecom and online services domains, by jointly modelling temporal dependencies and local trends in the data (Khattak et al., 2023; Li et al., 2025; Liu, Xia, et al., 2024). Additionally, an ensemble-based deep learning technique is introduced for integrating DTCN (Deep Temporal Context Network), RNN, and LSTM architectures, optimized using a Meta-heuristic Enhanced Dynamic Tuned Butterfly Optimization algorithm. Their approach achieved superior performance compared to traditional deep learning baselines, showcasing the promise of ensemble architectures in churn prediction tasks (Subramanian et al., 2024). In another direction, Sundaram et al. developed an LSTM-based representation learning framework tailored for sequential telecom data, optimizing predictive performance through deep feature abstraction (Sundaram et al., 2024). More recently, multimodal fusion approaches have emerged to integrate heterogeneous data sources for more holistic churn modelling. For example, Rudd et al. introduced a framework that combines voice sentiment analysis, customer financial literacy scores, and behavioural data to improve churn prediction in the financial services domain. Their work illustrates how fusing structured and unstructured modalities can enhance predictive performance, especially in settings with varied customer interaction formats (Rudd et al., 2023).

Also, A CNN-LSTM hybrid architecture, combined with ensemble learning strategies (e.g. bagging, boosting, stacking), is introduced for customer churn prediction in subscription-based settings. The model effectively captured both spatial and temporal customer behaviour patterns and achieved 98% accuracy, outperforming individual CNN and LSTM models. This integration of deep learning and ensemble techniques demonstrates a promising direction for robust, high-precision churn modelling in dynamic service environments (Nimma et al., 2025).

In some industries, such as the banking sector, churn modelling is often combined with Customer Lifetime Value (CLV) estimation to guide marketing and retention strategies. Unlike fast-churn domains like telecom or gaming, banking relies on long- term product relationships and multi-product customer behaviour. Recent work by Cowan et al. demonstrates a production-level CLV modelling framework developed for a major U.K. retail bank, combining segmentation, customer-specific transition probabilities, and a multi-year simulator. Their approach achieved a 43% improvement in out-of-time CLV prediction error over Markovian baselines, while remaining interpretable for targeted campaign use (Cowan et al., 2023).

CLV modelling has increasingly leveraged deep learning and meta-learning strategies to enhance predictive performance, robustness, and adaptability across diverse business contexts. One notable advancement comes from Gadgil et al., who proposed a meta-learning-based stacked regression framework for CLV prediction. Their model aggregates predictions from base learners such as bagging and boosting regressors using a neural meta-learner. Evaluated on a publicly available online retail dataset, the model significantly outperformed individual learners in terms of Mean Absolute Error and Root Mean Square Error, demonstrating improved generalization and accuracy in customer value estimation (Gadgil et al., 2023).

In a complementary direction, Wu et al. introduced a contrastive multi-view learning framework designed to address data sparsity and noise, i.e. challenges commonly encountered in real-world CLV scenarios. Their approach combines heterogeneous regressors and employs contrastive representation learning to fuse diverse customer behaviour signals. When tested on a large-scale mobile gaming dataset, the model achieved a 32.26% increase in total revenue impact compared to traditional single-view models. It was successfully deployed in Huawei's game centre, highlighting its practical utility in production environments (Wu et al., 2023).

Recently, an AI-powered CLV prediction system has been developed, tailored for real-time adaptation to behavioural changes in Business-to-Consumer retail. Their predictive engine integrates dynamic learning techniques to adjust to evolving customer preferences, producing accurate and actionable CLV scores that support personalized marketing and segmentation strategies. Case studies in the retail domain showed that this system enhanced customer retention and profitability through more precise targeting and campaign design (Egorenkov, 2024). These advances collectively reflect a shift towards more robust, scalable, and interpretable CLV modelling frameworks, powered by deep learning methods that address limitations in classical heuristics and static scoring systems.

While deep learning models show impressive accuracy, their black-box nature poses challenges for stakeholder adoption. There is an increasing demand for explainable deep learning frameworks tailored to churn analysis.

3.5. Evaluation metrics

The evaluation process of a predictive model involves the utilization of various metrics. Assessing the performance of ML models in all areas including the customer churn context is a crucial part of analysis, indicating the validity and effectiveness of the model performance. Considering the frequent imbalance in churn datasets, the selection of inappropriate evaluation metrics can lead to potentially misleading outcomes. In this part, we focus on commonly employed metrics for model performance evaluation. There are some popular key metrics that are leveraged for assessing the performance of churn prediction models namely, the area under the curve (AUC) of the receiver operating characteristic (ROC) curve, F1 score (Hasumoto & Goto, 2022) and the lift (Ahn et al., 2020; Alboukaey et al., 2020; Burez & Van den Poel, 2009). The AUC metric is seen as a general measure because it summarizes how well the model performs across all cut-off points (Alboukaey et al., 2020).

The ROC curve is created by plotting sensitivity on the y-axis and false positive rate on the x-axis. It's a robust measurement for classifiers, irrespective of class distribution or misclassification costs. In churn analysis scenarios, the x-axis represents non-churn cases incorrectly classified as churn (i.e. False Positive), and the y-axis represents churn cases correctly classified (i.e. True Positive) (Huang et al., 2012). A high AUC (close to 1) indicates accurate distinction between churn and non-churn customer characteristics (Hassouna et al., 2016; Runge et al., 2014). The AUC represents the aggregated performance measure and can be interpreted as the probability that the model correctly classifies a positive instance compared to a negative instance (Huang & Ling, 2005).

Some churn studies commonly utilize a performance metric known as the top 10% decile lift (Hassouna et al., 2016; Burez & Van den Poel, 2009; Risselada et al., 2010). The lift measure assesses observations or customers based on their predicted probability of being churners. Specifically, for the top 10% of customers considered the riskiest, the top decile lift is calculated as the ratio between the proportion of churners in this risky segment ($\pi_{10\%}$) and the overall proportion of churners in the validation set. The top decile lift technique is often preferred as it allows for the proportional allocation of marketing budgets to customers more likely to churn, as predicted by the churn prediction model (Burez & Van den Poel, 2009).

Some of the previous churn studies relied on accuracy as a performance metric, even in the presence of imbalanced churn datasets. However, using accuracy in such scenarios is not advisable, as it may generate unclear and unreliable results (Geiler et al., 2022). While AUC and lift are dominant, domain-specific metrics that reflect business Key Performance Indicators (KPIs), e.g. revenue loss prevention are rarely discussed. In practice, the choice of evaluation metric should be driven by the business context and associated costs of misclassification. For example, in the banking sector, a false positive (predicting a loyal customer as a churner) could result in unnecessary incentive offers, increasing operational costs. In contrast, a false negative (failing to identify a real churner) might lead to a direct loss of a high-value client, which is typically costlier. Therefore, precision becomes critical when targeting expensive retention actions, while recall is prioritized when missing actual churners has high financial consequences. In telecommunications, where customer bases are large, lift and AUC are widely used to rank potential churners and optimize campaign targeting within limited budgets. In online gaming, where customer value per user is lower, models might prioritize recall to retain as many users as possible. Thus, aligning metric selection with business impact, customer lifetime value (CLV), and budget constraints is essential for actionable churn prediction. Bridging the gap between statistical and business impact metrics could strengthen real-world applicability.

4. Cross-domain churn analysis: applied features and techniques

4.1. Data and features

Building on our discussion of churn definitions, we next examine the data-driven foundations of churn analysis, focusing on applied features and analytical techniques used across business domains. Data is essential to customer churn research because it provides a foundation for understanding and minimizing the probability of having customers left. Businesses can find trends and predictors of customer churn through using a variety of datasets, including customer-related data. Analysing this data and identifying the primary causes of churn behaviour (i.e. key features or churn predictors), provides a valuable technical benefit for formulating prediction models. These models enable businesses to predict potential churn occurrences and take preventative action (Hasumoto & Goto, 2022).

Many applications, including data analysis, intelligent systems, and large-scale computation, depend on machine learning. To be effective, ML workflows must have a good combination of data, features, and models (Brooks et al., 2015). Features and models lay between the raw data and the desired insights in the machine learning workflow. In that workflow, we choose the features in addition to the model. This approach has two joints, and selecting one affects the other. Appropriate features simplify the next modelling phase and increase the effectiveness of the resultant model in achieving the intended outcome. To obtain the same level of performance with bad features, a model may need to be considerably more complex (Zheng & Casari, 2018).

To deal with customer churn problem, different analytical approaches have been leveraged, using business-specific features. A feature is a metric that expresses raw data. They are individual measurable properties or variables of a data set used in machine learning. Naturally, the type of data that is provided would impact on the features. They can take on a variety of forms such as 'numerical features' (e.g. age, income, or temperature), 'categorical features' (e.g. gender, country, and occupation), 'textual features' (e.g. Bag-of-Words, TF-IDF, Word Embeddings, and text length or word count), 'temporal features' (e.g. date and time), 'geospatial features' (e.g. Latitude and longitude, and distance or proximity), and 'transactional features' (e.g. products prices). Features are also influenced by the model, with certain models better suited to particular feature types than others. The ideal features should be simple for the model to process and pertinent to the task. The number of features is also crucial. The model won't be able to complete the task at hand if there aren't enough informative features. The model will cost more and be more challenging to train if there are too many features or if the majority are useless

(Zheng & Casari, 2018). The process of creating the most suitable features for the task, the model, and the data is known as feature engineering. Feature engineering is a crucial technique in machine learning due to the challenges associated with raw data and the need to extract meaningful information for effective model training and prediction. It refers to the process of transforming raw data into a format that is suitable for machine learning algorithms to effectively learn patterns and make accurate predictions (Zheng & Casari, 2018; Xie et al., 2023).

It involves creating new features, modifying or transforming existing ones to enhance the predictive power of machine learning models by providing them with more informative and representative features (Reddy et al., 2020). It could be considered a generic term for any tasks used between gathering raw data and starting model training, and the selection of the appropriate approach is dependent on the *ML* model (Xie et al., 2023). Comparing to the non-contractual businesses, researchers in contractual settings focus more on identifying churn predictors and feature engineering process, as well as exploring the available data (Ascarza et al., 2018). Nevertheless, its significance is evident in both situations, as it is essential to improving the prediction model's performance as well as creating plans for retention and monitoring (Hasumoto & Goto, 2022). In this way, a wide range of machine learning techniques can be used using structured features, which are readily available. Majority of previous churn studies applied structured data in their analysis (Hoppner et al., 2020; Maldonado et al., 2020; Wang et al., 2020), while a few considered both structured and unstructured data together (Vo et al., 2021; Slof et al., 2021; Amiri & Daume, 2015).

Customer data typically includes a range of valuable information that provides insights into the dynamics of the customer–company relationship. Used data in churn prediction studies are categorized into two primary groups, i.e. structured and unstructured data. Structured or numerical data typically refers to organized, tabular information, which can include various types of quantitative data (Mitrovi´c et al., 2018). On the other hand, Unstructured data are those kept in their original format and tend to have a qualitative character. They need to be processed and handled by specialists for machine learning algorithms to use them. Email correspondence with customers, phone talks, and chat interactions with support staff are a few instances of unstructured data.

Slof et al. study proved the positive contribution of topic variables that were extracted out of textual data on model performance (Slof et al., 2021). Similar to this, Vo et al. used text mining and NLP techniques to predict churn (Vo et al., 2021). The driving force in their research was the existence of the low correlation between structured variables and the target variable. Models that included unstructured variables performed better than those that only included structured variables, according to both experiments (Slof et al., 2021; Vo et al., 2021). The research emphasizes how important information-rich content is in consumer-business communications, leading to promising predictive outcomes. Churn-related features generally could be categorized in various groups, as follows:

- Product-Usage features:
  This features group includes data reflecting how customers behave and engage with the product or service, e.g. product use, duration of the usage, specific product features accessed, and RFM (Recency, Frequency, Monetary)-based features. Features related to how recently and how frequently customers engage with a product or service have been shown to be crucial in predicting customer churn (Tamaddoni et al., 2016;

Coussement & De Bock, 2013; Migueis et al., 2012). The number of outgoing calls a customer makes each day is an instance of a product use feature. Usually, these features are collected daily, weekly, or monthly and then included to churn models. Usage features aggregated on a daily basis perform better than those aggregated on a weekly or monthly basis, according to Alboukaey et al. (Alboukaey et al., 2020).

Call detail records (CDR) is a popular data in telecommunication domain. CDR contains information about how customers use telecom products or services. On the other hand, it is indicative of customers' communication behaviour as well as social interaction, such as the frequency of calls, duration of calls, and types of calls made (voice or non-voice), and the times when calls are made. It includes information such as 'Caller and Callee Information', 'Call Start and End Times', 'Call Duration', 'Call Duration', 'Call Type', 'Call Location', 'Call Quality', 'Call Charges'. Analysing CDR can provide insights into the communication patterns of customers and how they relate to churn behaviour (Kim et al., 2014).

On the other hand, the recency feature helps to notice how recently a customer has made a purchase. This helps identify if a customer is active or inactive. Frequency is indicative of how often a customer makes a purchase. This helps identify how engaged a customer is. Monetary features refer to the financial aspects of customer behaviour and interactions, i.e. the money a customer spends on purchases. This helps identify high-value customers (Fader et al., 2005). RFM-based features are mostly used in marketing and customer relationship management to analyse and categorize customers. It helps in identifying and targeting specific customer segments for personalized marketing strategies (Alboukaey et al., 2020).

Leveraging RFM-based models are a widely adopted approach in research for describing customer interactions and behaviour. It is a straightforward but effective method that allows the measurement of customer interactions based on recency, frequency, and monetary aspects. In their study, by optimizing a profitability function, Ho¨ppner et al. (Hoppner et al., 2020) and Maldonado et al. (Maldonado et al., 2020) combined monetary features with more structured features to predict churn. These studies emphasized how crucial it is for researchers to focus on finding the most financially advantageous model in addition to the most effective one.

- Support Service and Satisfaction Features:
  This group of features should be considered carefully in the customer churn analysis. Those data reflect the customers' interactions with the company's support services, such as reporting damaged items, poor quality, returns of the goods, and refunds (Tamaddoni et al., 2016), inquiry volumes, resolution times, and customer feedback or satisfaction ratings (Agrawal et al., 2018; Kim et al., 2014; Hoppner et al., 2020). Collectively, these dimensions form a comprehensive understanding of customers, enabling companies to tailor their services and support to meet customer needs and enhance overall satisfaction. However, much past research has identified satisfaction and service quality as main factors contributing to customer churn (Zeithaml et al., 1996; Sifa et al., 2014).

- Demographic Features:
  This group outlines a customer's profile, encompassing factors such as their geographic location, age, and marital status.

- Social Interaction Features:
  This group includes customer's social behaviour in social network features, i.e. social interactions. They are defined by social network attributes within the customer group. In this network, every customer is represented as a node, and the links between nodes are represented as edges. For example, in the context of telecommunications in which the CDR data is available (Kim et al., 2014), an edge could be the total number of calls that two users have received inbound. Metrics such as centrality can also be extracted from a social network once it has been built and used as features for churn prediction (Calzada-Infante et al., 2020).

  Another type of features in this group are Social Network Analysis (SNA) features that are accessible from enhanced call networks, introduced by Mitrovi´c et al. (2021). Enhanced call networks are communication networks where additional information such as call forwarding, call waiting, voicemail, and other value-added services, beyond basic call details are incorporated. As a result of their studies, Ahmad et al. (2019) and Oentaryo et al. (2012) emphasized that adding SNA features improves classifier performance. However, the primary limitation of SNA features is their computing cost (Mitrovic & De Weerdt, 2020; Postigo-Boix & Melus-Moreno, 2018).

Today, availability of large volumes of log data offers a valuable resource for customer churn analysis. Log data refers to records or traces of activities and events generated by computer systems, applications, or devices during their operation. In the context of churn prediction, log data may include information about user interactions, transactions, errors, or any relevant events that can help analyse and understand user behaviour and predict the likelihood of churn. Thus, models predicting churn with log data can apply to Internet services across different industries. This is a growing trend that industries like gaming and telecommunications, with abundant log data and easy access to customer information (Vo et al., 2021), are increasingly using deep learning with big data. In financial and insurance sectors, where log data is limited and customer information changes less, statistical approaches and traditional machine learning models, including survival analysis, are more common (Ahn et al., 2020; Mavri & Ioannou, 2008).

The majority of contemporary churn prediction analyses are done utilizing log data (Lee, Jang, et al., 2018; Kim et al., 2017; Vo et al., 2021). Log data refers to a record of events, actions, or transactions within a system, application, or network.[2] In other words, the trace data left over after using On-line services is known as a log. As a result, a variety of businesses can employ online services and churn prediction models that were created utilizing log data (Ahn et al., 2020; Lee, Jang, et al., 2018). Transaction logs (e.g. purchases, subscription renewals, or cancellations), communication logs (i.e. used to keep track of email messages, phone calls, and other forms of communication), and login and activity logs (i.e. customer logins and their subsequent activities) are some instances of log data. Focusing on some of the main industries (i.e. telecommunication, banking, and insurance) affected by customer churn problems, in the following parts, churn definition in different industries and the way various features are considered to be applied in customer churn analysis are reviewed.

Due to its importance, customer churn research has been promoted in various service businesses. Different businesses have their nature and characteristics. Because of that, the churn problem has been studied separately in each of them, based on different churn measurement criteria. Previous customer churn works are expanded in industries such as telecommunication (e.g. mobile services), internet services, games, insurance, finance, management, music streaming (Ahn et al., 2020; Burez & Van den Poel, 2007). Tables 3 and 4 show some of the previous churn studies in multiple domains, considering applied features from different aspects. Despite extensive use of structured data, unstructured sources (e.g. emails, chats) remain underutilized. Their inclusion may offer richer behavioural insights, especially when fused with traditional features. Recent studies

Table 3. Feature comparison table, showing some of the previous churn studies in multiple domains, considering applied features from different aspects.

Year	Domain	Type of Features	Text	Env.	Temporal behaviour	T-series	Log data	Ref.
2025	Gaming	Demog., Prdct-Usg (RFM inc.)	×	×	✓ (Daily)	✓	✓	Li et al. (2025)
2025	Telecom	Prdct-Usg, Demog.	✓	×	×	×	✓	Shahabikargar et al. (2025)
2025	Telecom	Prdct-Usg, Demog.	×	×	×	×	×	Nimma et al. (2025)
2025	Telecom	Prdct-Usg	×	×	✓ (Monthly)	×	×	Asif et al. (2025)
2024	Gaming	Prdct-Usg	✓	×	✓ (Daily)	✓	×	Abdul-Rahman et al. (2024)
2024	Multiple Industries	Usage, Demog.	×	×	×	×	×	Liu, Jiang, et al. (2024)
2024	Telecom	Prdct-Usg, Demog., Support S.	×	×	✓ (Annual)	×	×	Wagh et al. (2024)
2024	Telecom	Prdct-Usg	×	×	✓ (Daily)	✓	✓	Subramanian et al. (2024)
2024	Banking	Demog., Prdct-Usg (RFM inc.)	×	×	✓ (Monthly)	×	×	Vu (2024)
2024	Telecom	Prdct-Usg, Demog.	×	×	✓ (Monthly)	×	×	Saha et al. (2024)
2024	Telecom	CRM-derived Features	×	×	×	×	×	Sagming et al. (2024)
2023	E-Commerce	Demog., Prdct-Usg (RFM inc.)	✓	×	✓ (Monthly)	×	✓	Shobana et al. (2023)
2023	Telecom	Prdct-Usg, Demog.	×	×	✓ (Monthly)	×	×	Khattak et al. (2023)
2023	Banking and Insurance	Prdct-Usg, Demog.	×	×	✓ (Monthly)	✓	×	Kate et al. (2023)
2022	Banking	Prdct-Usg, Demog.	×	×	✓ (Daily)	×	×	Bharathi et al. (2022)
2022	Platform, Businesses	Prdct-Usg (RFM inc.)	×	×	✓ (Daily)	✓	×	Hasumoto and Goto (2022)
2022	Social Networking	Demog., Social Int. (i.e.Product Review)	×	×	×	×	×	Zhang et al. (2022)
2022	Telecom	Prdct-Usg, Demog.	×	×	×	×	×	Lalwani et al. (2022)
2021	Telecom	Prdct-Usg, Demog., Support S.	✓	✓	✓ (Monthly)	×	✓	Slof et al. (2021)
2021	Banking	Prdct-Usg, Demog., Support S.	✓	×	×	×	×	Vo et al. (2021)
2020	Banking	Prdct-Usg, Demog., Support S.	✓	×	×	×	×	De Caigny et al. (2020)
2020	Telecom	Prdct-Usg (RFM inc.)	×	×	✓ (Daily)	✓	×	Alboukaey et al. (2020)
2019	Telecom	Social Int. (CDR), Prdct-Usg, Demog.	×	×	✓ (Daily)	×	×	Ahmad et al. (2019)
2019	Gaming	Prdct-Usg, Demog.	×	×	✓ (Daily)	✓	✓	Kristensen and Burelli (2019)

Notes: It indicates whether 'Unstructured textual' features are used or not. It also shows if any features indicative of 'Environmental' factors (Env.) such as competitor prices and stock market information are included in the corresponding research or not. It also shows on which 'Temporal' basis (Temp.) (i.e. daily-based or monthly-based behaviour) the data is analysed, whether 'Time-series' as well as 'Log data' features are leveraged or not in the corresponding study.

Table 4. Feature comparison table, showing some of the previous churn studies in multiple domains, considering applied features from different aspects (continue).

Year	Domain	Type of Features	Text	Env.	Temporal behaviour	T-series	Log data	Ref.
2018	Social Networking	Social Int., (messaging app)	×	×	✓ (Daily)	✓	×	Yang et al. (2018)
2018	Gaming	Prdct-Usg, Demog.	×	×	✓ (Daily)	×	✓	Liu et al. (2018)
2018	Telecom	Prdct-Usg, Demog., Support S.	×	×	×	×	×	Hoppner et al. (2020)
2018	Telecom	Demog., Prdct-Usg	×	×	×	×	×	Agrawal et al. (2018)
2017	Telecom	Prdct-Usg, Demog.	×	×	×	×	×	Umayaparvathi and Iyakutti (2017)
2017	Insurance	Prdct-Usg, Demog.	×	×	✓ (Daily)	×	×	Zhang et al. (2017)
2017	Gaming	Prdct-Usg	×	×	✓ (Daily)	×	✓	Milosevic et al. (2017)
2017	Gaming	Prdct-Usg	×	×	✓ (Daily)	×	✓	Kim et al. (2017)
2016	Retailing	Demog., Prdct-Usg (RFM inc.), Support S.	×	×	×	×	✓	Tamaddoni et al. (2016)
2015	Gaming	Prdct-Usg	×	×	✓ (Biweekly)	×	✓	Xie et al. (2015)
2015	Telecom	Demog., Social Int. (microblogs)	✓	✓	×	×	×	Amiri and Daume (2015)
2015	Telecom	Prdct-Usg	×	×	×	×	×	Vafeiadis et al. (2015)
2015	Banking/ Insurance	Prdct-Usg, Demog.	×	×	✓ (Daily)	×	×	Sundarkumar and Ravi (2015)
2014	Insurance	Prdct-Usg, Demog.	×	×	✓ (Monthly)	✓	×	Gunther et al. (2014)
2014	Telecom	Social Int. (CDR), Demog., Prdct-Usg, Support S.	×	×	×	×	×	Kim et al. (2014)
2014	Telecom	Social Int. (CDR), Demog., Prdct-Usg	×	×	✓ (Monthly)	×	×	Verbeke et al. (2014)
2014	Gaming	Prdct-Usg, Feedback	×	×	✓ (Daily)	×	✓	Grandprey-Shores et al. (2014)
2014	Gaming	Prdct-Usg, Demog.	×	×	✓ (Daily)	✓	✓	Runge et al. (2014)
2014	Retailing	Prdct-Usg (RFM inc.)	×	×	×	×	✓	Jahromi et al. (2014)
2014	Banking	Prdct-Usg(RFM inc.)	×	×	✓ (Monthly)	✓	×	Ali and Arıturk (2014)
2012	Insurance	Prdct-Usg, Demog.	×	×	✓ (Monthly)	×	×	Soeini and Rodpysh (2012)
2010	Telecom/ Insurance	Prdct-Usg, Demog.	×	✓	×	×	×	Risselada et al. (2010)
2009	Banking	Prdct-Usg, Demog.	×	×	✓ (Daily)	×	×	Xie et al. (2009)
2008	Newspaper	Prdct-Usg, Demog., Support S.	✓	×	×	×	×	Coussement and Van den Poel (2008)
2008	Banking	Support S.	×	×	×	×	×	Mavri and Ioannou (2008)
2007	Telecom	Prdct-Usg, Demog.	×	×	×	×	×	Chu et al. (2007)
2004	Banking	Prdct-Usg	×	✓	×	×	×	Lariviere and Van den Poel (2004)
2004	Banking	Prdct-Usg, Demog.	×	✓	×	×	×	Van den Poel and Lariviere (2004)
2003	Banking	Prdct-Usg	×	×	✓ (Daily)	×	×	Chiang et al. (2003)

have started to explore the integration of Natural Language Processing (NLP) and multi-modal frameworks for churn prediction. For instance, Slof et al. (2021) applied topic modelling using Latent Dirichlet Allocation on customer support emails, improving churn model performance when combined with structured data. Similarly, Vo et al. (2021) used TF-IDF and sentiment analysis to extract features from support chats, demonstrating enhanced churn prediction over structured-only baselines. In a more advanced application, transformer-based models such as BERT have been used to capture semantic nuances from customer complaints and reviews (e.g. Rudd et al., 2023), and combined with structured features in multimodal deep learning pipelines. These approaches illustrate the potential of multimodal fusion, where textual, behavioural, and demographic data are combined in architectures such as attention-based CNN-LSTM networks or deep Autoencoders + Multi Layer Perceptron hybrids to achieve superior performance in predicting churn.

4.2. Churn in various domains

4.2.1. Telecommunications

Most past churn studies have focused on the telecommunications sector (telecom), being the most impacted by customer churn.[3] Churn in telecom services refers to the change of service provider from one to another (Ahn et al., 2020; Dahiya & Bhatia, 2015). Churn, as described by Alboukaey et al., is the occurrence of a customer ceasing revenue-generating activities inside the 30-day prediction window after they were active throughout the 90-day observation window (Alboukaey et al., 2020). Generally, churn in telecom is defined as when a customer is left the service provider. In other words, the termination of all services that a customer has been receiving is interpreted as a churn event in the corresponding telecom or service operator (Hoppner et al., 2020; Agrawal et al., 2018; Lalwani et al., 2022).

There are several types of telecommunication service categories, such as telephone (landline and wireless services), mobile, internet, cable or satellite television. Among them, mobile services have received much more attention and most previous studies have focused on it (Ribeiro et al., 2024). Costs to acquire customers are significant for these services. Therefore, it would be of great help if customer churn is avoided and proper incentives are offered (Ahn et al., 2020; Hung et al., 2006). Churn rate is a particularly useful measurement in the telecommunications industry. As most customers have multiple options from which to choose, the churn rate helps a company determine how it is measuring up to its competitors. For example, if one out of every 20 subscribers to a high-speed internet service terminated their subscriptions within a year, the annual churn rate for that internet provider would be 5%.

Tables 5 and 6 are illustrating some of the previous churn studies in multiple domains, considering applied models, the evaluation metrics as well as the best performing model. As shown on this table, various research has studied customer churn problems in the telecom domain. Those studies leveraged plenty of traditional models such as Decision trees (Dahiya & Bhatia, 2015; Bahnsen et al., 2015), Random forests (Colot et al., 2021), SVM (Huang et al., 2012), logistic regression (Verbraken et al., 2014), Ensemble (Baumann et al., 2015), and Boosting (Vafeiadis et al., 2015) methods as well as deep learning models (Mishra & Reddy, 2017; Umayaparvathi & Iyakutti, 2017). Table A1 includes the abbreviations and the corresponding phrases.

Based on a recent Systematic Literature Review (SLR)regarding customer churn in telecommunication industry, age, gender, satisfaction, switching costs and barriers, and service quality are the key factors influencing customer's propensity to quit a service provider and switch to a competitor. Service quality indirectly impacts customer churn through satisfaction. Socio-demographic variables like age and gender show mixed impacts on customer's propensity to churn. Switching costs and barriers, such as loyalty contracts and complex service plans, play a significant role in reducing customer

Table 5. Model comparison table, showing some of the previous churn studies in multiple domains, considering applied models, the evaluation metrics (Eval. M) as well as the best performing model.

Year	Domain	Best Model	Eval. M	Benchmark Model(s)	Ref.
2025	Gaming	BiLSTM + attention	Acc, AUC, Lift	XGB, LSTM, BiLSTM	Li et al. (2025)
2025	Telecom	XGB	Prec., Rec., F1, AUC	LightGBM, AdaBoost, Tabnet, MLP, Gradient Boosting	Shahabikargar et al. (2025)
2025	Telecom	CNN-LSTM Hybrid	Acc., Prec., Rec., F1	CNN, LSTM	Nimma et al. (2025)
2025	Telecom	TriBoost (i.e. an ensemble)	F1, Acc., Prci., Recall, AUC	DT, LR, GNB, Ridge Cl., SGD	Asif et al. (2025)
2024	Multiple Industries	XGB	Acc., Prci., Recall, F1, AUC	RF, LR, SVM, GNB	Liu, Jiang, et al. (2024)
2024	Telecom	RF + OverS (SMOTE)	F1, Acc., Prci., Recall	DT,Survival Analysis	Wagh et al. (2024)
2024	Telecom	Stacked Ensemble(CNN + RNN + LSTM)	F1, Acc., Prci., Recall	CNN, RNN, LSTM, SVM, RF, XGB	Subramanian et al. (2024)
2024	Telecom	RF	Acc., Prci., Recall, F1, AUC, Gini Coe.	LR, DT, SVM, KNN, XGB, NB, MLP, Ens. M	Sagming et al. (2024)
2024	Banking	Stacked Ensemble (DNNs, LR)	F1, Acc., Prci., Recall	XGB, KNN, SimpleRNN, LSTM	Vu (2024)
2024	Telecom	ChurnNet (i.e. 1D CNN)	Acc., Prci., Recall, F1, AUC, POF, MCC	LR, SVM, RF, MLP, CNN, LSTM, GRU, Ensemble	Saha et al. (2024)
2023	E-Commerce	SVM + ANN	Acc., Lift, Prci., Hit/Coverage Rate	SVM, ANN	Shobana et al. (2023)
2023	Telecom	BiLSTM-CNN	Acc., Prci., Recall, F1	SVM, XGB, DT, KNN	Khattak et al. (2023)
2023	Banking and Insurance	RF + OverS, GANs + UnderS, DT	AUC, F1	RF, SVM, LR, MLP, LGBM	Kate et al. (2023)
2022	Banking	Ens. M, ExtraTrees	F1	Traditional Models	Bharathi et al. (2022)
2022	Telecom	Boosting	AUC	Traditional Models	Lalwani et al. (2022)
2022	Platform Businesses	RF	F1, Acc., Prci., Recall	RF + RFM	Hasumoto and Goto (2022)
2022	Social Networking	CFChurn (a GCN-based model)	AUC, Acc.	DiffM, Emb., GCN, GAT, RF, XGB, Survival Model	Zhang et al. (2022)
2021	Telecom	Survival analysis	Lift	Survival analysis with/ without Text-driven Feature	Slof et al. (2021)
2021	Banking	Ens. M, Transfer Learning	AUC	XGB, LR, GNB, RF	Vo et al. (2021)
2020	Banking	CNN	AUC, Lift	LR	De Caigny et al. (2020)
2020	Telecom	LSTM	AUC, F1, Lift	RF, LR	Alboukaey et al. (2020)
2020	Banking	RF, RF + OverS(GANs)	F1	RF + OverS(SMOTE)	Li and Xie (2020)
2019	Telecom	XGB	AUC	DT, RF, GB	Ahmad et al. (2019)
2019	Gaming	LSTM	AUC	RF, NN	Kristensen and Burelli (2019)
2018	Gaming	GNN	AUC, Recall	LR, DT, RF, SVM	Liu et al. (2018)
2018	Telecom	Proposed ProfTree	AUC, F1, a PDM	DT	Hoppner et al. (2020)
2018	Telecom	Deep Learning	Acc.	No-Comparison	Agrawal et al. (2018)
2018	Social Networking	Clustering + LSTM + Att. M	Acc., Prci., Recall	LR, RF	Yang et al. (2018)

Table 6. Model comparison table, showing some of the previous churn studies in multiple domains, considering applied models, the evaluation metrics (Eval. M) as well as the best performing model (continue).

Year	Domain	Best Model	Eval. M	Benchmark Model(s)	Ref.
2017	Telecom	CNN	Acc.	SVM, RF	Umayaparvathi and Iyakutti (2017)
2017	Insurance	Deep Learning + LR, DSM	AUC, F1	CNN, LSTM, SGD, LDA QDA, GNB, Ada.B, RF, GTB	Zhang et al. (2017)
2017	Gaming	Boosting	AUC, F1	LR, NB, DT, RF	Milosevic et al. (2017)
2017	Gaming	Boosting, LSTM	AUC	RF, LR, CNN	Kim et al. (2017)
2016	Retailing	Boosting	Lift, a PDM	LR, SVM, Stat. M	Tamaddoni et al. (2016)
2015	Telecom	SVM, LR	F1	N-gram Model	Amiri and Daume (2015)
2015	Telecom	SVM, Ada.B	F1	Traditional Models	Vafeiadis et al. (2015)
2015	Banking and Insurance	GMDH, KRNN + OCSVM	AUC, Sensitivity	Traditional Models	Sundarkumar and Ravi (2015)
2015	Gaming	DT	AUC, Prci., Recall	LR, SVM	Xie et al. (2015)
2014	Telecom	LR, Graph	AUC, Lift	No-Comparison	Kim et al. (2014)
2014	Telecom	Graph, Traditional Models	a PDM	No-Comparison	Verbeke et al. (2014)
2014	Insurance	LR, GAMs	ROC, TP, FP	No-Comparison	Gunther et al. (2014)
2014	Banking	LR/DT	AUC, Lift	LR + OverS, DT + OverS, Stat. M (Cox Regression)	Ali and Arıturk (2014)
2014	Gaming	NN + HMM	AUC	LR, DT, SVM	Runge et al. (2014)
2014	Gaming	LR	Stat. M	No-Comparison	Grandprey-Shores et al. (2014)
2014	Retailing	Boosting	AUC, Lift, a PDM	DT, DT + CSL, Boosting, LR	Jahromi et al. (2014)
2012	Insurance	CART, Clustering (Kmeans)	ROC curve, Lift	No-Comparison	Soeini and Rodpysh (2012)
2010	Telecom and Insurance	DT + Bagging	Lift, Gini Coe.	LR, LR + Bagging, DT	Risselada et al. (2010)
2009	Banking	RF + CSL	Lift	DT, FR, NN	Xie et al. (2009)
2008	Banking	Survival analysis	Stat. M	Author-designed	Mavri and Ioannou (2008)
2008	Newspaper	LR + with TDF	AUC, Lift	LR + without TDF	Coussement and Van den Poel (2008)
2007	Telecom	DT, Clustering	Acc.	No other models	Chu et al. (2007)
2004	Banking	Survival analysis	Stat. M	Author-designed	Van den Poel and Lariviere (2004)
2004	Banking	Survival analysis	Stat. M	Author-designed	Lariviere and Van den Poel (2004)
2003	Banking	Rule-based modelling	Rule-based modelling	No-Comparison	Chiang et al. (2003)

churn. Customers with more tenure are less likely to switch to another service provider. Companies should correctly allocate customers to a tariff (attributes of pricing plans) that matches their consumption profile. Bundled services are effective in reducing customer churn. Competitive environment, advertising messages, and promotional campaigns of competitors impact churn and switching intentions (Ribeiro et al., 2024).

4.2.2. Banking and insurance

Customer churn analysis is critical to customer retention strategies, market share growth, and cost-effective acquisition of new customers in the present financial and banking sector (Vo et al., 2021; Charandabi & Ghanadiof, 2022). Compared to other industries, the banking sector struggles with relatively low customer retention, highlighting significant challenges in maintaining long-term client relationships. In response, banks are increasingly prioritizing the use of digital innovations and data-driven approaches to enhance customer experience and satisfaction (Singh et al., 2024).

All industries are affected by the fast growth of service providers, with the banking sector experiencing a particularly large increase. Customers have a wide range of options when deciding where to put their money into the banking industry. Therefore, keeping active customer involvement levels and managing customer churn are major concerns for many banks (Rahman & Kumar, 2020). In banking, an individual who cancels all of his accounts and ceases doing work with the bank is referred to as a churned customer. Customer churn can have severe effects on the banking business and the loss of funds and revenues. As a result, a key component of customer-oriented retention, is identification of customers who exhibit a high propensity to quit the bank as well as predicting customers who would leave the company in the future (Karvana et al., 2019). Maintaining loyal customers and continuously monitoring their satisfaction and areas of improvement are essential aspects for industries such as banking, insurance, and retail (Khan et al., 2019; Bharathi et al., 2022).

Machine learning has been incorporated into customer data analytics to predict customer churn. Despite successful applications of ML models into customer churn analysis, existing approaches primarily use only structured data, e.g. demographics and product usage data. On the other hand, data mining with unstructured data, e.g. customer interaction and call chats have not been adequately leveraged. Vo et al. proposed a customer churn prediction model utilizing both structured and unstructured customer data (i.e. the verbal information exchanged during telephone conversations), in a banking scenario (Vo et al., 2021). The results indicated the performance superiority of the proposed ensemble approach, leading to the generation of meaningful insights leveraging Explainable Artificial Intelligence (XAI), in addition to personality traits analysis and customer segmentation. On their research, they leveraged content mining combined with natural language processing (NLP) with focusing on more finely granular customer affective statues, using Text-driven Features (i.e. by leveraging LIWC (Linguistic inquiry and word count)) (Pennebaker et al., 2001), TF-IDF (Scherer et al., 2015), Word2Vec (Goldberg & Levy, 2014), Personality Traits (Goldberg, 2013) in addition to Product-Usage (e.g. account balance and investment decisions) as well as demographic features. Further, they applied the advanced transfer learning approach to the call logs with the similar spoken text nature to build a better churn prediction model which utilizes personality features (Vo et al., 2021).

In this context of insurance companies, when a customer terminates all their policies, whether due to changing insurance companies or no longer requiring insurance coverage, this is called customer churn. Insurance companies can no longer depend on a consistent customer network due to accessible insurance provider switching rules. In many countries, insurance regulations allow customers to terminate policies anytime, not just on the due date. With numerous competitors and more aware customers, retaining customers is crucial. The cost of acquiring new customers is significantly higher than keeping existing ones, and even a slight rise in retention rates can substantially boost premium revenue. Therefore, preventing customer churn plays a vital role in insurance industries' success (Gunther et al., 2014; Zhang et al., 2017).

Soeini et al. study yielded valuable insights, including the following recommendations. Firstly, offering electronic services to speed up the registration process. Secondly, provide customers with detailed information about the services and benefits offered by their policies. Thirdly, expand the service offers, even at higher price points, with a focus on catering to special customers. Fourthly, furnishing customers with guidance on how to claim loss recovery payments, thereby enhancing their satisfaction. Lastly, introduce a system where the insurer directly covers some customers' insurance bills to prevent them from incurring losses (Soeini & Rodpysh, 2012).

In today's competitive landscape, insurance providers are progressively adopting omnichannel and personalized engagement strategies, aiming to deliver a cohesive and integrated customer experience across multiple touchpoints, ranging from physical branches and call centres to digital platforms like websites and mobile applications(Gerea et al., 2021).

4.2.3. Healthcare industry

Health insurance churn is a well-documented phenomenon within the healthcare industry, comparable to patterns observed in other sectors. Sommers et al. (2016) emphasize that life events such as job changes, income reductions, marriage, or childbirth frequently trigger transitions in coverage eligibility. These shifts are particularly prevalent among low-income individuals, whose incomes often vary seasonally or unpredictably, placing them near critical eligibility thresholds for public insurance programmes. Such financial instability increases the likelihood of frequent coverage loss and re-enrolment. Furthermore, affordability remains a significant determinant of churn; when confronted with high insurance premiums, many individuals may opt to forgo health coverage in order to prioritize essential expenses like rent, food, or transportation.

Building on this, Hill and Jacobs (2025) examined changes in coverage stability following the implementation of the Affordable Care Act (ACA). While they observed modest improvements in average coverage durations, they found that insurance churn remained a persistent issue, especially among individuals in the private insurance market. The authors emphasized that fluctuations in income and employment status continued to disrupt coverage, particularly for those near subsidy eligibility thresholds. They also highlighted administrative hurdles, such as reapplication processes and plan switching, further exacerbated instability in coverage continuity.

Similarly, Kong et al. (2022) investigated turnover in zero-premium status among Marketplace health insurance plans, particularly those targeting low-income populations. Their findings revealed that seemingly small shifts in plan pricing or subsidy structures could cause a loss of zero-premium eligibility, prompting enrolees to either switch plans or drop coverage altogether. This volatility undermines long-term retention and presents a substantial challenge to maintaining continuous care, especially for vulnerable populations who may not have the financial literacy or timely awareness to respond to such changes. Together, these studies underscore that insurance churn in the healthcare sector is not only a product of personal life changes but also deeply influenced by policy design, market dynamics, and system-level complexity. Ensuring stable insurance coverage for at-risk groups requires targeted policy interventions aimed at smoothing income-related eligibility transitions, reducing administrative friction, and improving affordability through sustained subsidy structures.

4.2.4. Internet-based services

User acquisition and retention are crucial factors that determine the survival and growth of various Internet-based services i.e. online applications, including social media platforms, blogs, online communities, news sites, games, and e-commerce websites (De Vries et al., 2017; Zhang et al., 2021). With the increasing number of online applications, competition for users has intensified, making it more challenging to acquire new users (Xu et al., 2021). As a result, predicting user churn, or the likelihood of users discontinuing their use of an application, has become a critical focus on this Internet-based services (Yang et al., 2018; Khodadadi et al., 2020). The advancement of digital technology has facilitated the accumulation of vast datasets that contain information about customer behaviour on internet-based services. Consequently, there is a growing interest in research that utilizes cutting-edge data-mining techniques to analyse these datasets and gain insights into customer behaviour (Ngai et al., 2009). On the other hand, The rise of new business models has heightened the demand for more sophisticated modelling approaches, in this area. However, implementing feature engineering to choose and modify critical features in such business models present challenges (Hasumoto & Goto, 2022).

Previous churn studies in contractual settings have been extended to internet services. For example, research on music streaming (Nimmagadda et al., 2017) and Software-as-a-Service (SaaS) businesses (Ge et al., 2017) shows promising applications of the XGBoost technique for churn analysis. On the other hand, various churn studies have been conducted in non- contractual settings, typically focused on retail businesses (Tamaddoni et al., 2016; Dingli et al., 2017). There is now a growing interest in exploring new sectors, such as platform businesses, social media and networking services (Calzada-Infante et al., 2020), and online gaming (Kristensen & Burelli, 2019; Guitart et al., 2018).

Platform Businesses are included in the category of non-contractual internet servicing businesses. They have experienced significant growth over the past decade. Platform businesses are typically defined as entities that provide a marketplace codes and foundation for producers and consumers (Van Alstyne et al., 2016) or facilitate interactions between different groups of consumers and complementors (Boudreau & Hagiu, 2009). Examples of platform businesses include Amazon (connecting sellers and buyers), Google (connecting advertisers and users), Apple (connecting app developers and users), Uber (connecting drivers and riders), Airbnb (connecting hosts and guests), and YouTube (connecting content creators and viewers).

A key feature of platform businesses is their ability to generate a network effect, where increased demand from one side of the platform stimulates demand from the other side. As more buyers join the platform looking for products, sellers are motivated to join and offer their products to reach more potential customers. This, in turn, increases the variety and availability of products on the platform, making it more attractive to buyers. Customers' purchasing behaviour, including their purchase cycle and average purchase amount, is influenced by both individual characteristics and the characteristics of the stores and items sold. While transaction records do not contain product-level information, they include details such as the time of purchase, amount spent, and store of purchase. This information enables the analysis of purchasing behaviour based on store categories and time-series trends to identify potential churn. In platform businesses, customers can explicitly close their accounts on the platform, but most customers churn by simply ceasing to make purchases, leaving their accounts open. This non-contractual relationship between consumers and the business further complicates the analysis of churn behaviour (Hasumoto & Goto, 2022; Van Alstyne et al., 2016; Boudreau & Hagiu, 2009). **Online Social Application** is another internet-based services that could face customer churn. As social media has become more widespread, many applications have incorporated social features like comments and friend sharing to improve user experience. Some applications have even built their entire platforms on social media. In this rapidly expanding environment, social influence has emerged as a key factor contributing to customer churn, where users stop using a service (Nitzan & Libai, 2011). Consequently, researchers have been exploring ways to model the impact of social influence on user churn in order to enhance the accuracy of churn prediction. Their efforts can be broadly categorized into two approaches. The first approach involves modelling social network structures to represent social influence. For instance, Ahmad et al. (2019) utilize social network analysis to extract network-based features for machine learning models. The second approach treats user churn as a diffusion process, assuming that if user A stops using a service after their friend B does, user A was influenced by user B. Researchers in this category often use diffusion models to simulate how social influence spreads (Zhang et al., 2012).

On the other hand, there is still an existing challenge in this context, that is whether the correlation between a user's churn and their social factors implies a causal relationship, where the social factors directly influence the user's decision. The challenges and gaps in existing approaches are twofold. First, using correlational methods to solve causal problems can introduce non-causal noise, limiting the performance of deep learning methods for churn prediction. Second, existing methods struggle to provide explanations for their predictions, hindering the ability to tailor campaigns for user retention (Zhang et al., 2022).

Online Retailing domain was also considered in churn prediction studies. Most of the previous studies investigated churn in settings with a Business to Consumer (B2C) context. Since there are limited churn studies in the companies with the Business to Business (B2B) context, Jahromi et al. proposed a data-mining approach to fill the gap. In their experiments, they leveraged a major Australian online Fast Moving Consumer Goods (FMCG) retailer dataset, including RFM features. Authored defined churn as customers being inactive in the second half of the year while active in the first half. The authors compared several modelling techniques, including CART and boosting, with logistic regression, and find that boosting performs best in terms of AUC and lift metrics. They then used the churn probabilities from the best model to develop a profit-maximizing retention campaign (Jahromi et al., 2014).

Gaming is one of the most popular domains that attracted the attention of many churn studies. In the last ten years, internet-connected devices, particularly smartphones, have become easily available. This has led to a shift in how people use these devices, with more frequent but shorter interactions. As a result, casual games have become incredibly popular. Simultaneously, developing these games has become simpler due to improved development tools. This procedure leads to an intense competition in the industry, rapid customer churn (Kim et al., 2017) as well as increase in Customer Acquisition Cost (CAC) (Goldani & Goldani, 2018; Xia & Jin, 2008). Therefore, user acquisition and retention becomes key challenges in the industry (Kim et al., 2017), while the importance of predicting and preventing churn grows at the same time.

Online games can be categorized in various ways, e.g. AAA games, Massively Multi-player Online Games (MMOGs), Free-to-Play (F2P) games, Mobile games, and eSports games, being games played competitively. AAA games, such as 'Destiny', are high-budget titles with advanced graphics and complex game-play mechanics. Persistent online games or MMOGs create virtual worlds where players interact simultaneously, with examples like World of Warcraft. Free-to-Play games, like League of Legends, and Fortnite, are free to download but generate revenue through in-game purchases. Mobile games, such as Candy Crush Saga, are designed for smartphones and often monetized through ads or in-app purchases. Lastly, eSports games are played competitively at a professional level in organized tournaments, exemplified by titles like League of Legends. Each category has its own unique characteristics and revenue models, contributing to the diverse landscape of modern gaming (Tamassia et al., 2016).

The gaming industry extensively employs ML for churn studies due to the vast amount of log data available (Lee, Jang, et al., 2018; Hadiji et al., 2014; Yang et al., 2020). In this domain, various KPIs are used to measure and assess the success, performance, and engagement of games (Hadiji et al., 2014; Yang et al., 2020). Churn analysis in the gaming industry utilizes several baseline features such as 'the playtime distribution data', 'daily total time spent', 'daily standard deviation of playtime distribution', 'hourly standard deviation of playtime distribution'. Another possible feature is the combined feature of 'total time spent, last login day, and number of time slots played', being designed based on the RFM model (Fader et al., 2005). These features help understand player behaviour and predict churn (Yang et al., 2020).

In previous studies, as shown in Table 7, churn prediction in the game industry has used different approaches and models like binary predictions, survival analysis, and the Cox model (Viljanen et al., 2016). These studies have also used various features, such as playtime, login frequency, player in-game status, and player activity like purchases or other KPIs (Jeon et al., 2017; Lee, Jang, et al., 2018; Lee, Kim, et al., 2018). Although it is not a very popular method in churn prediction, Tamassia and Rothenbuehler applied Hidden Markov Model (HMM) on their churn studies in the gaming domain, i.e. AAA games and Freemium games, respectively (Tamassia et al., 2016; Rothenbuehler et al., 2015). They both compared the performance of HMM with other ML models such as LR and NN and conclude the superiority of the HMM. An HMM is a statistical tool for describing real-world processes with observable signals. HMMs rely on a hidden Markov chain, providing advantages for expert systems due to low storage and computational needs, chosen for its suitability with the time series data.

5. Conclusion: insights, limitations and future directions

5.1. Comparative evaluation

Given the variation in data availability, customer behaviour, and churn definitions across industries, it is valuable to analyse methodological trends within the context of specific domains. Our comparative tables, Tables 1 and 2, which compare previous churn analysis studies across different domains; Tables 3 and 4, which summarize the types of features employed; and Tables 5 and 6, which outline the models and evaluation metrics used, highlight how the choice of features, models, and performance criteria is inherently

Table 7. Churn prediction model comparison on gaming domain.

Year	Category of gaming	Best model	Benchmarked models	
2025	F2P	BiLSTM + attention	XGB, LightGBM, LSTM, BiLSTM	Li et al. (2025)
2024	AAA game	BiLSTM-CNN	RF, LR, LSTM	Abdul-Rahman et al. (2024)
2023	F2P	CMLTV (Contrastive Multi-view)	MLP, RF, XGB	Wu et al. (2023)
2023	F2P	New distance measure + PAM + RF	Gower distance + PAM Clus., Kamila Clus.	Periˇsi´c and Pahor (2023)
2023	F2P & MMORPG	PCATlr	RF, PCAT	Weiss and Vilenchik (2023)
2019	F2P	LSTM + Dynamic and Static data	RF, NN	Kristensen and Burelli (2019)
2018	MMORPGs	RF	RF, XGB, GBM	Lee, Kim, et al. (2018)
2018	F2P	LSTM	Survival Ensembles	Guitart et al. (2018)
2018	F2P	GNN	LR, DT, RF, SVM	Liu et al. (2018)
2017	F2P	RF	DT	Jeon et al. (2017)
2017	F2P	Survival Ensembles	Cox regression, Kaplan-Meier	Bertens et al. (2017)
2017	F2P	Boosting	LR, NB, DT, RF	Milosevic et al. (2017)
2017	F2P	Boosting/LSTM	RF, LR, CNN	Kim et al. (2017)
2016	AAA games	HMM	LR, RF, Boosting, DT, NN, NB, Bagging	Tamassia et al. (2016)
2016	AAA game	Ensemble of KNN + RF + GBC + Ada.B	KNN, RF, GBC, Ada.B	Sifa et al. (2016)
2016	F2P	Ensemble	LR, RF, SVM	Drachen et al. (2016)
2016	F2P	Survival Ensembles	Cox regression	Perianez et al. (2016)
2015	F2P	DT	LR, SVM	Xie et al. (2015)
2015	F2P	RF	DT, SVM	Sifa et al. (2015)
2015	F2P	HMM	LR, NN, SVM	Rothenbuehler et al. (2015)
2014	F2P	LR	No Comparison	Grandprey-Shores et al. (2014)
2014	F2P	NN + HMM	LR, DT, SVM	Runge et al. (2014)
2014	F2P	DT	LR, NN, NB	Hadiji et al. (2014)

shaped by industry characteristics and business objectives. For instance, the telecommunications and gaming industries benefit from abundant log data and high-frequency customer interactions, which support the application of deep learning architectures and graph-based models. In contrast, the banking and insurance sectors typically rely on structured, transactional data, making traditional machine learning and statistical approaches, such as logistic regression and survival analysis, more suitable.

Moreover, churn definitions are domain-specific, with contractual churn in telecom and banking enabling clear binary or time-to-event modelling, whereas non-contractual churn in e-commerce or gaming mostly requires sequence modelling techniques such as LSTM and BiLSTM (Li et al., 2025; Liu, Xia, et al., 2024; Vatani et al., 2025). Industry-specific constraints also shape feature engineering practices, for example, gaming domains prioritize real-time behavioural logs (Costa et al., 2024), while financial services emphasize RFM features, demographics, and credit profiles. Finally, evaluation metrics are aligned with business goals, with telecom studies often using AUC and lift to guide marketing interventions, whereas financial applications focus more on precision and recall to avoid costly false positives. These distinctions underscore the importance of contextualizing churn analysis methodologies based on the specific operational and data environments of each industry.

5.2. Limitations

Despite the breadth of research surveyed, several limitations persist across methodologies and applications. In this section, we discuss key challenges that continue to hinder progress in churn analysis. These ongoing challenges require further exploration and refinement of current methodologies.

Lack of Customer-generated Data Analysis: Customer-generated data, i.e. unstructured textual data, could provide valuable insights into customer behaviour and emotions (Vo et al., 2021; Slof et al., 2021; De & Prabu, 2022). It includes chat logs, reviews, social media posts, and feedback. Although this type of data could be effective indicators of churn (Vo et al., 2021; Slof et al., 2021), As shown in Table 3, integrating these unstructured variables within the churn prediction framework remains under-explored. The research conducted by Vo et al. and Slof et al. also supports the idea that.

Imbalanced Churn Data: One of the primary limitations in customer churn prediction lies in dealing with imbalanced data. Churn datasets often exhibit class imbalance, where the number of churners significantly differs from the number of non-churners. Typically, a dataset is considered class imbalanced if the difference between churn classes is 35% or more (De & Prabu, 2022). This poses a significant barrier as standard classification algorithms may struggle to effectively learn from such datasets. To address the imbalanced data issue, various methods have been proposed. SMOTE (Anil Kumar & Ravi, 2008), under-sampling (Smith et al., 2014), over-sampling (Han et al., 2005; He et al., 2008). These methods can be applied individually or combined to balance the dataset (LemaÃŽtre et al., 2017; Ahmad et al., 2019).

Limited Consideration of Environmental Factors: Due to their significance, these factors, e.g. competitors' offerings, should be vastly considered on churn studies (Lariviere & Van den Poel, 2004). As shown in Table 3, previous studies have largely overlooked the impact of environmental factors on churn analysis. Some of the probable reasons might be (1) the corresponding data might not be readily available or easily accessible, (2) even when such data is available, it may not be of sufficient quality or granularity for meaningful analysis. Understanding and incorporating these environmental factors into churn analysis models can provide valuable insights for businesses seeking to improve customer retention strategies.

Limited Application of Time Series Data: Most of previous works have looked at churn as a static prediction problem (Lalwani et al., 2022; Bharathi et al., 2022; Vo et al., 2021), while a limited number of research, as demonstrated on Table 3, explored churn analysis from a dynamic perspective (Kate et al., 2023; Alboukaey et al., 2020; Hasumoto & Goto, 2022). Time series data analysis allows us to capture temporal patterns and trends in customer churn behaviour over time. By analysing historical churn data, It is possible to identify recurring patterns, seasonality effects, and long- term trends that may influence customer retention (Kate et al., 2023; Alboukaey et al., 2020).

Limited Clustering Approaches: Although applying the clustering approach proved to be helpful in churn analysis (Soeini & Rodpysh, 2012; Yang et al., 2018; Periˇsi´c & Pahor, 2023), as shown in Table 5, limited number of previous studies explored its efficacy on churn analysis. Leveraging clustering could add a layer of granularity to churn analysis, enabling companies to tailor their strategies to different customer segments and improve the effectiveness of their churn prevention efforts (Soeini & Rodpysh, 2012; Yang et al., 2018).

Table 8. A demonstration of available churn-related datasets on different domains (all links were last checked on April 2024).

Dataset's name	Domain	Link to Access	Ref.
South Korean Telecom Operator	Telcom	https://www.kaggle.com/datasets/mahreen/sato2015	✓ Hoppner et al. (2020)
IBM-telco-customer-churn	Telcom	https://github.com/IBM/telco-customer-churn-on-icp4d/tree/master/data	✓ Agrawal et al. (2018), Nimma et al. (2025)
Iranian churn	Telecom	https://archive.ics.uci.edu/datasets?search = CHURN	×
Nigeria Telecoms Churn	Telecom	https://www.kaggle.com/c/dsntelecomschurn2018/data	×
Churn Telco Europa	Telecom	https://www.kaggle.com/datasets/raumonsa11/churn-telco-europa	×
Churn Prediction with Text and Interpretability	Telecom	https://github.com/aws-samples/churn-prediction-with-text-and-interpretability	×
Mobile-churn-data	Telecom	https://www.kaggle.com/datasets/dimitaryanev/mobilechurndataxlsx	×
Cell2cell	Telecom	https://www.kaggle.com/datasets/jpacse/telecom-churn-new-cell2cell-dataset	✓ Umayaparvathi and Iyakutti (2017)
Churn modelling dataset	Banking	https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling	×
Bank Customer Churn Prediction	Banking	https://www.kaggle.com/churn-prediction-using-ann-dl/data	×
Zenodo-Dodge the Mud	Gaming	https://zenodo.org/records/166035	✓ Kim et al. (2017)
Zenodo-Undisclosed	Gaming	https://zenodo.org/records/166035	✓ Kim et al. (2017)
World-of-Warcraft	Gaming	https://github.com/Customer-Churn-in-World-of-Warcraft/data	×
Newspaper churn	Newspaper	https://www.kaggle.com/datasets/andieminogue/newspaper-churn	×
KDD cup 2009	CRM	https://www.kdd.org/kdd-cup/view/kdd-cup-2009/Data	×
WSDM – KKBox's Churn Prediction Challenge	Music	https://www.kaggle.com/c/kkbox-churn-prediction-challenge/data	×
MembershipWoes	Club	https://www.kaggle.com/datasets/varshapandey/assignment-data	×

Limited Number of Publicly Available Churn Datasets: Numerous research present findings based on confidential datasets, making it hard to validate or replicate their results (De & Prabu, 2022). Through this research, we managed to identify some of the available datasets for customer churn analysis, as shown in the Table 8. Most of the available datasets are related to the telecom domain, and only six datasets were used in the reviewed papers in this work.

5.3. Future works

Based on the extensive background churn studies on this work, several promising directions for future research are proposed here. Future research in customer churn prediction can focus on several promising directions. Key areas include integrating customer feedback, leveraging advanced analytics techniques, dynamic modelling, ethical considerations, and enhancing data quality. By incorporating human and AI feedback, crowdsourcing, and advanced reinforcement learning, customer feedback can significantly improve retention strategies.

Advanced analytics, such as predictive analytics, segmentation, embedding methods, and Retrieval Augmented Generation (RAG), can refine models and insights. Dynamic modelling through real-time data analysis and continuous learning allows for timely detection and proactive interventions. Ethical considerations, including data privacy regulations and bias mitigation, are crucial for responsible and fair analysis. Finally, enhancing data quality through improved cleaning and collection methods ensures the reliability and accuracy of insights, ultimately leading to more effective retention strategies and better business outcomes. These proposed directions highlight the potential of future churn research in addressing some of the limitations and challenges in the customer churn area.

Notes

1. Pre-paid mobile telephony refers to a mobile phone service where users pay for the service in advance by purchasing credits.
2. https://www.sumologic.com/glossary/log-file/.
3. https://techsee.me/resources/surveys/state-of-customer-churn-telecom-survey-report/.

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Appendix

Table A1 includes the abbreviations used in this study and the corresponding phrases:

Table A1. Abbreviation table.

Abbreviation	Phrase	Abbreviation	Phrase
Acc.	Accuracy	B2B	Business to Business
B2C	Business to Consumer	CAC	Customer Acquisition Cost
CLV	Customer Lifetime Value	CSL	Cost-sensitive Learning
DT	Decision Tree	DSM	Deep Shallow Model
DiffM	Diffusion-based Models	Demog.	Demographic
ELM	Extreme Learning Machine	Emb.	Embeddings
Ens. M	Ensemble Model	Extra Trees	Extremely Randomized Trees
F2P	Free-to-Play	FMCG	Fast Moving Consumer Goods
GAI	Generative Artificial Intelligence	GAN	Generative Adversarial Networks
GBM	Gradient Boosting Machines	GAMs	Generalized Additive Models
GNB	Gaussian Naive Bayes	Gini Coe.	Gini Coefficient
GMDH	Group Method of Data Handling	GNN	Graph Neural Networks
GHSOM	Growing Hierarchical Self-Organizing Map	GTB	Gradient Tree Boosting
HMM	Hidden Markov Model	IR/SR	Infection Rule/Stoppng Rule
KNN	K-Nearest Neighbours	kRNN	k-Reverse Nearest Neighbourhood
LDA	Linear Discriminant Analysis	LGBM	Light Gradient Boosting Machine
LLR	Logistic Lasso Regression	LR	Logistic Regression
LSTM	Long Short-Term Memory	MLP	MultiLayer Perceptron
MMOGs	Massively Multiplayer Online Games	MMORPGs	Massively Multiplayer Online Role-Playing Games
NB	Naive Bayes	NBD	Negative Binomial Distributions
NN	Neural Network	OCSVM	One-Class Support Vector Machine
OverS	Over Sampling	PAM	Partitioning Around Medoids
PCAT	PCA Trajectories	PCATLR	Linear Regression Variant of PCAT
PDM	Profit Driven Metric	PNN	Probabilistic Neural Network
Prci.	Precision	Prdct-Usg	Product-Usage
QDA	Quadratic Discriminant Analysis	RAG	Retrieval-Augmented Generation
RF	Random Forest	RFM	Recency, Frequency, Monetary
RNN	Recurrent Neural Network	SGD	Stochastic Gradient Descent
SPA	Spreading Activation Technique	Stat. M	Statistical modelling
SVM	Support Vector Machine	TDF	Text-Driven Feature
UnderS	Under Sampling	XGB	XGBoost
Social Int.	Social Interactions	RFM inc.	RFM included
Support S.	Support Services	Ridge cl.	Ridge Classifier
SGD	Stochastic Gradient Descent	ENN	Edited Nearest Neighbours
POF	Probability of False Alarm	MCC	Matthews Correlation Coefficient
Ens. M	Ensemble Model	Gini Coe.	Gini Coefficient